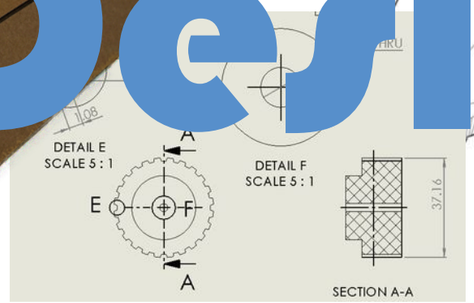
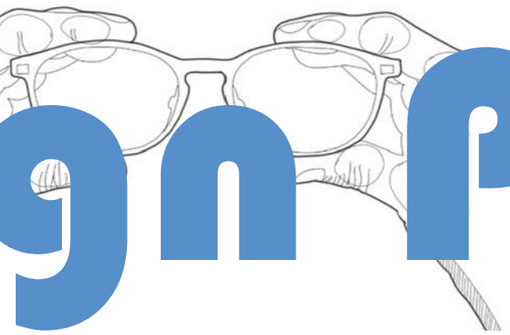
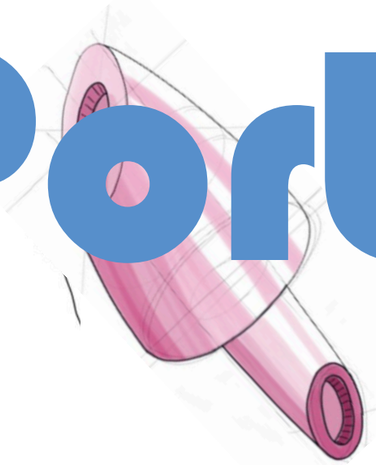
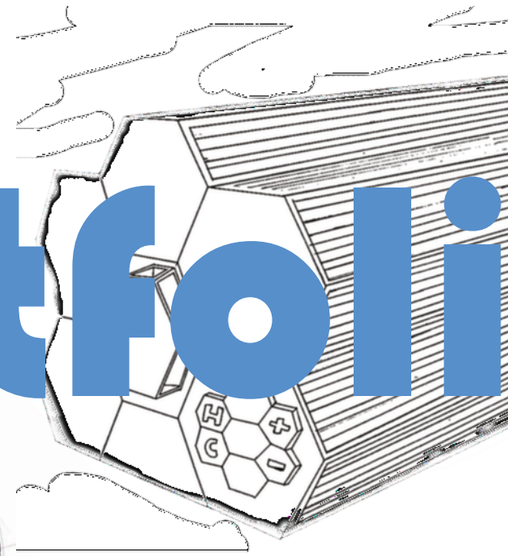
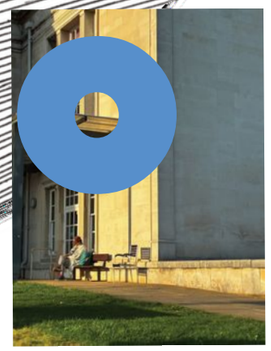












Design Portfolio

Rachel Zinkin

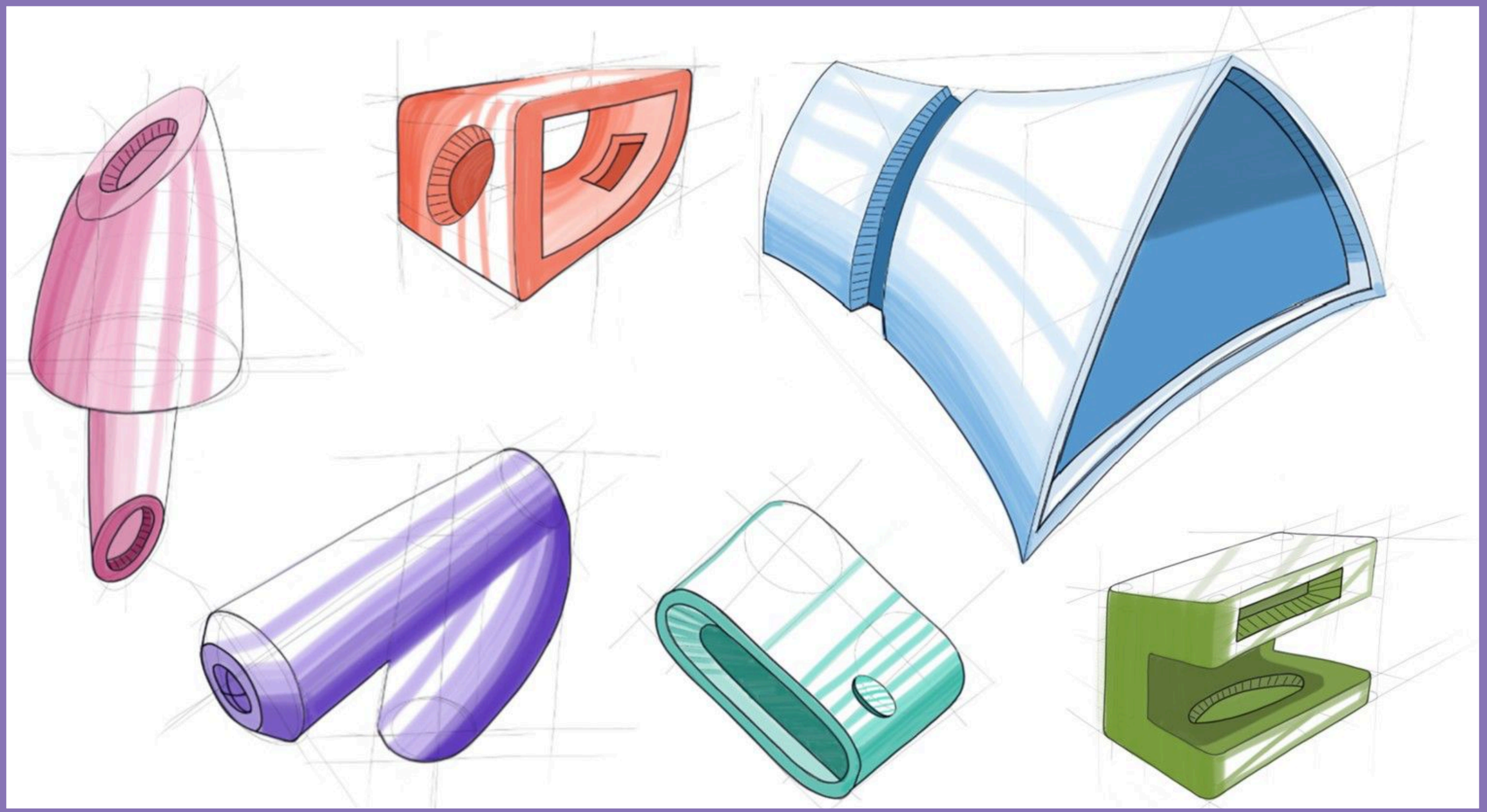
Drawing Pages 3-6

Project work Pages 7-26

Technical skills Pages 27-31

Drawing skills

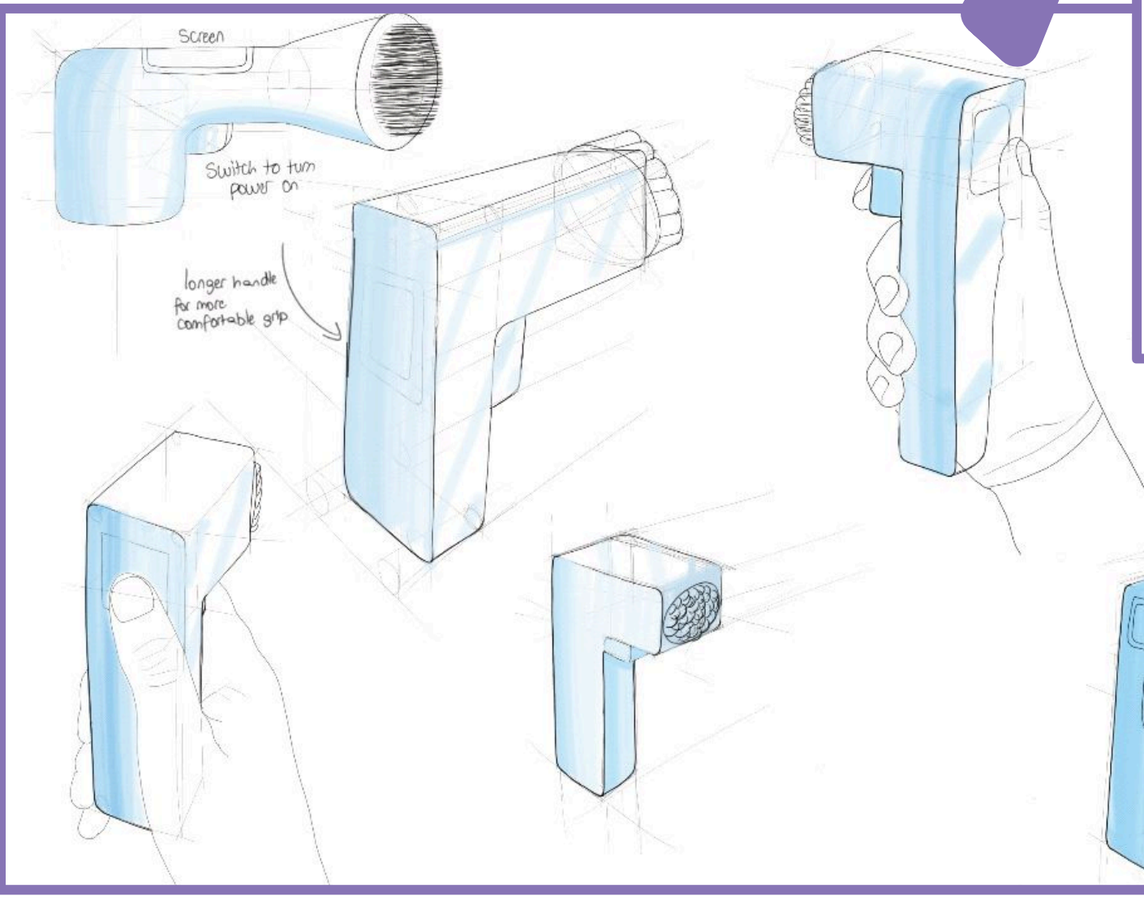
Drawing - digital drawing: level 6+8 drawings with colour

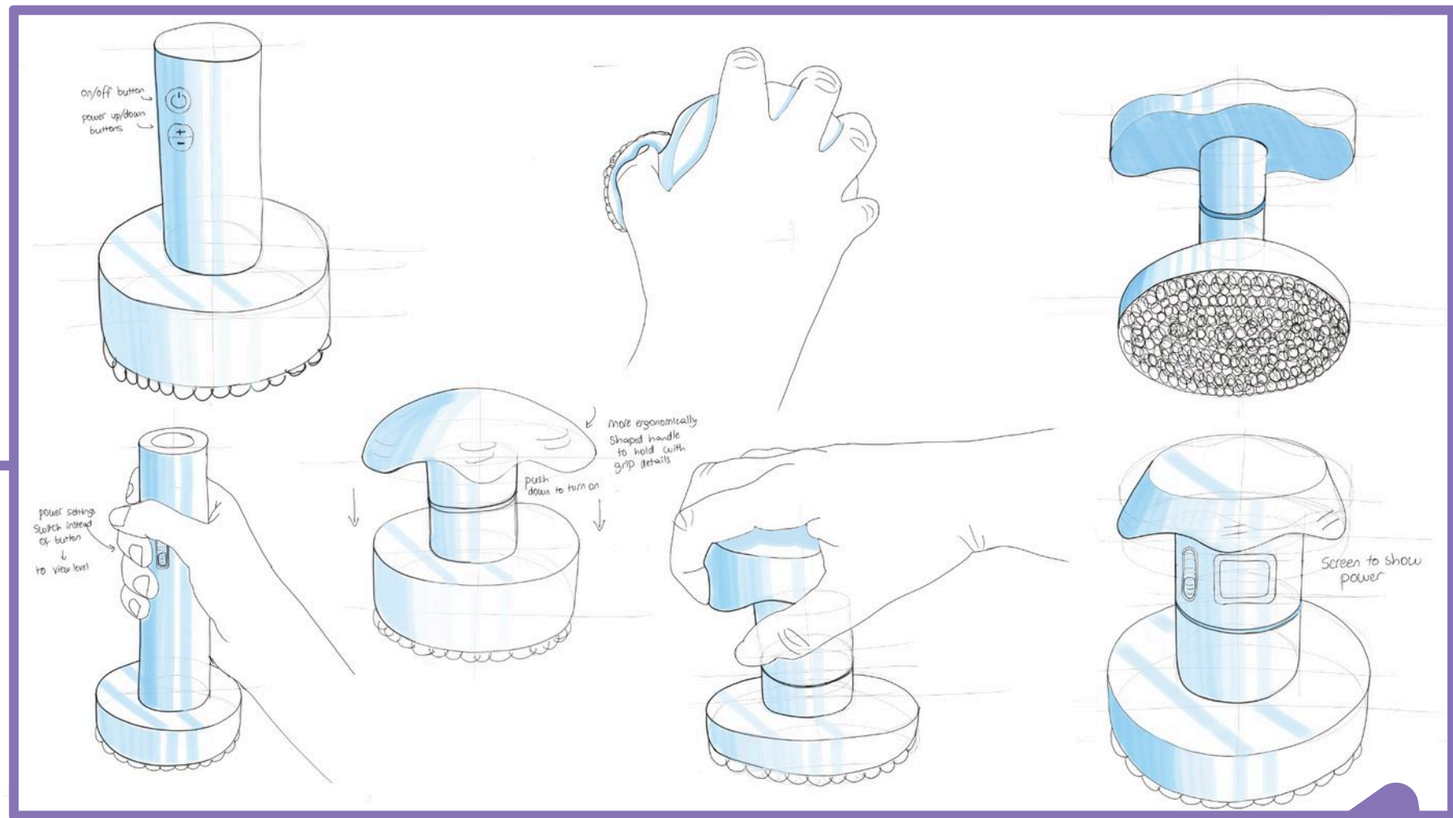


Digital drawing using marker to depict light and shade.

Drawing - digital drawing: concept development of scrub brush

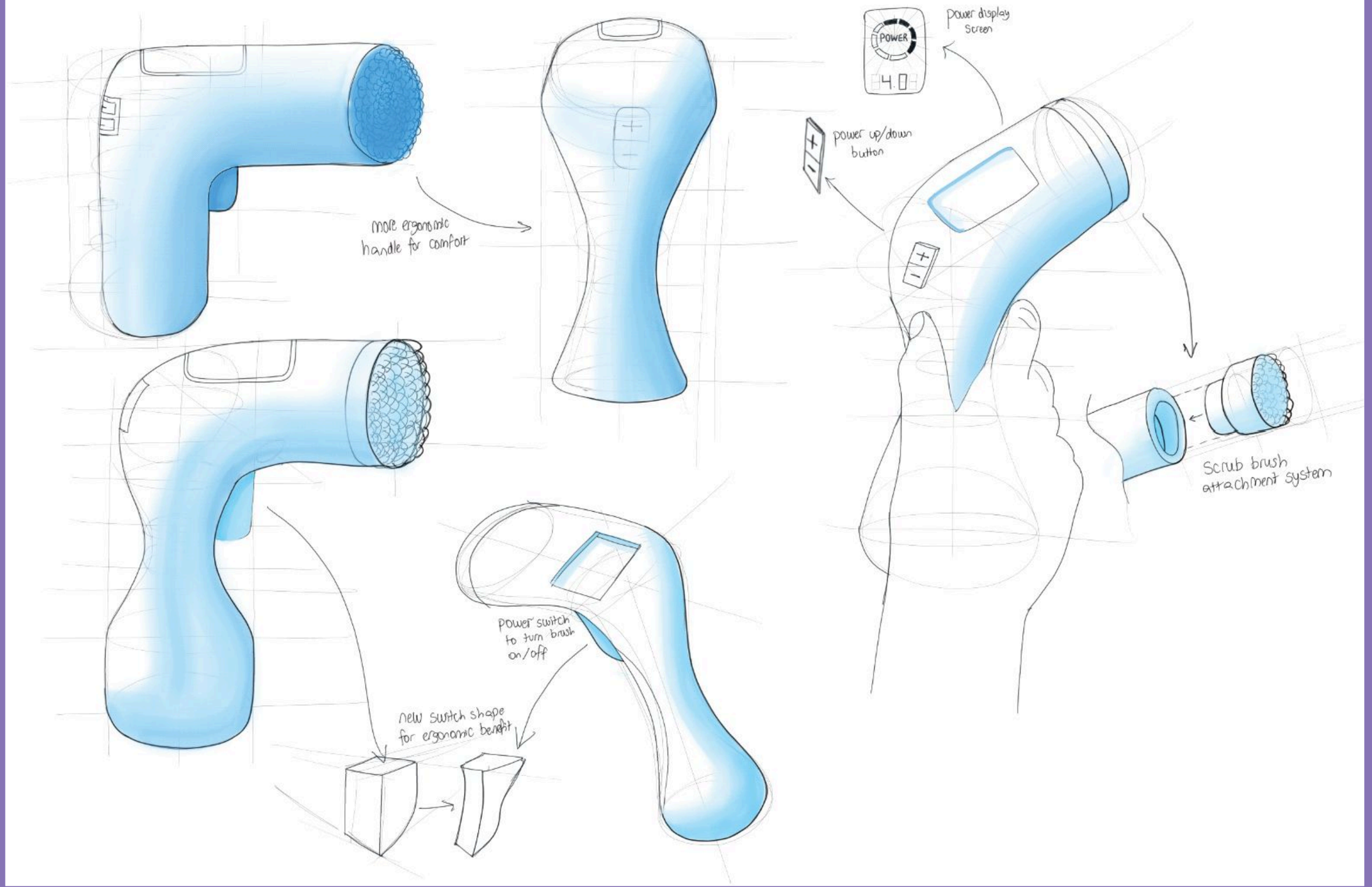
Concept development 1





Concept development 2

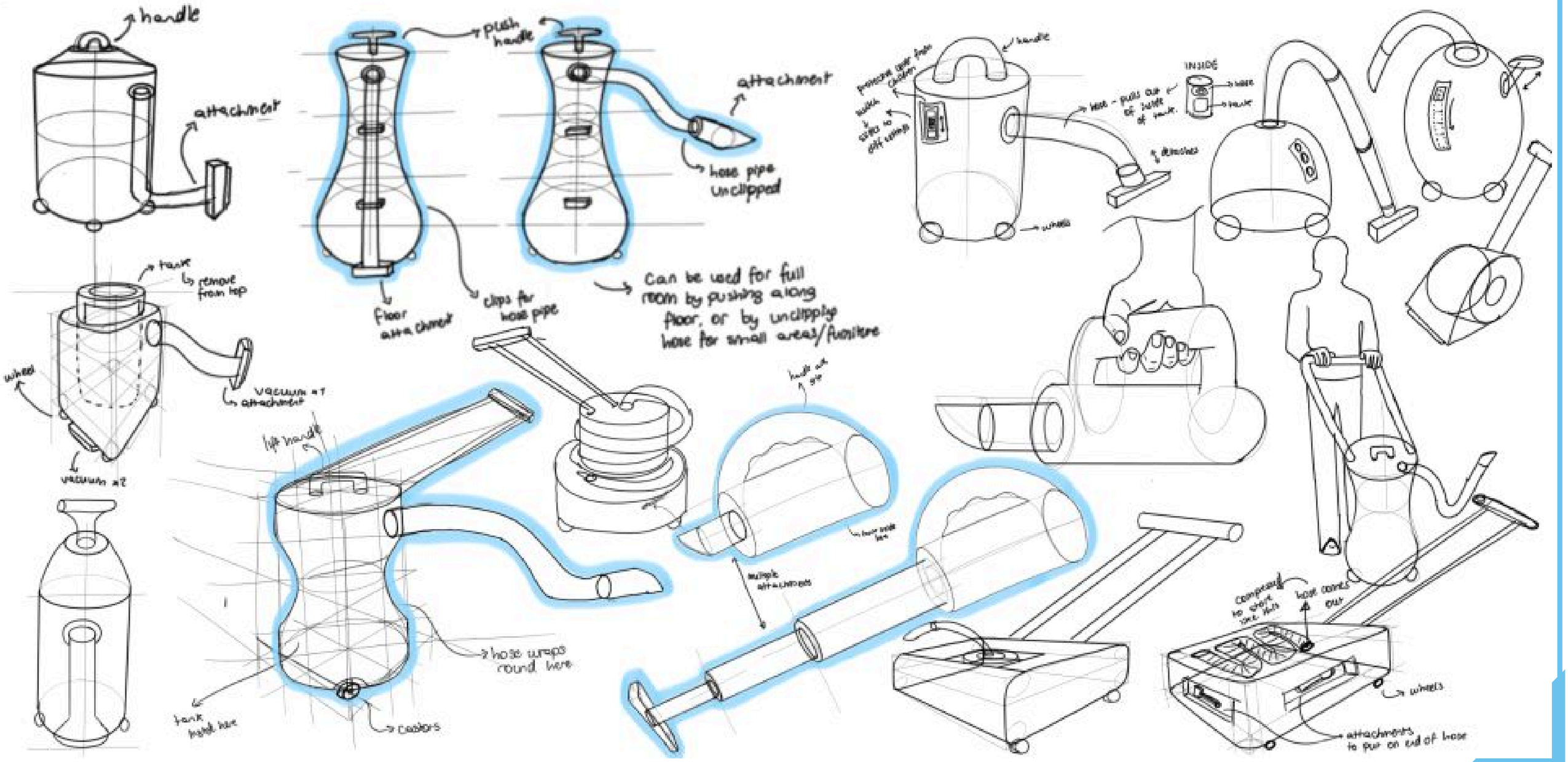
Drawing - digital drawing: refinement of scrub brush design



Project work

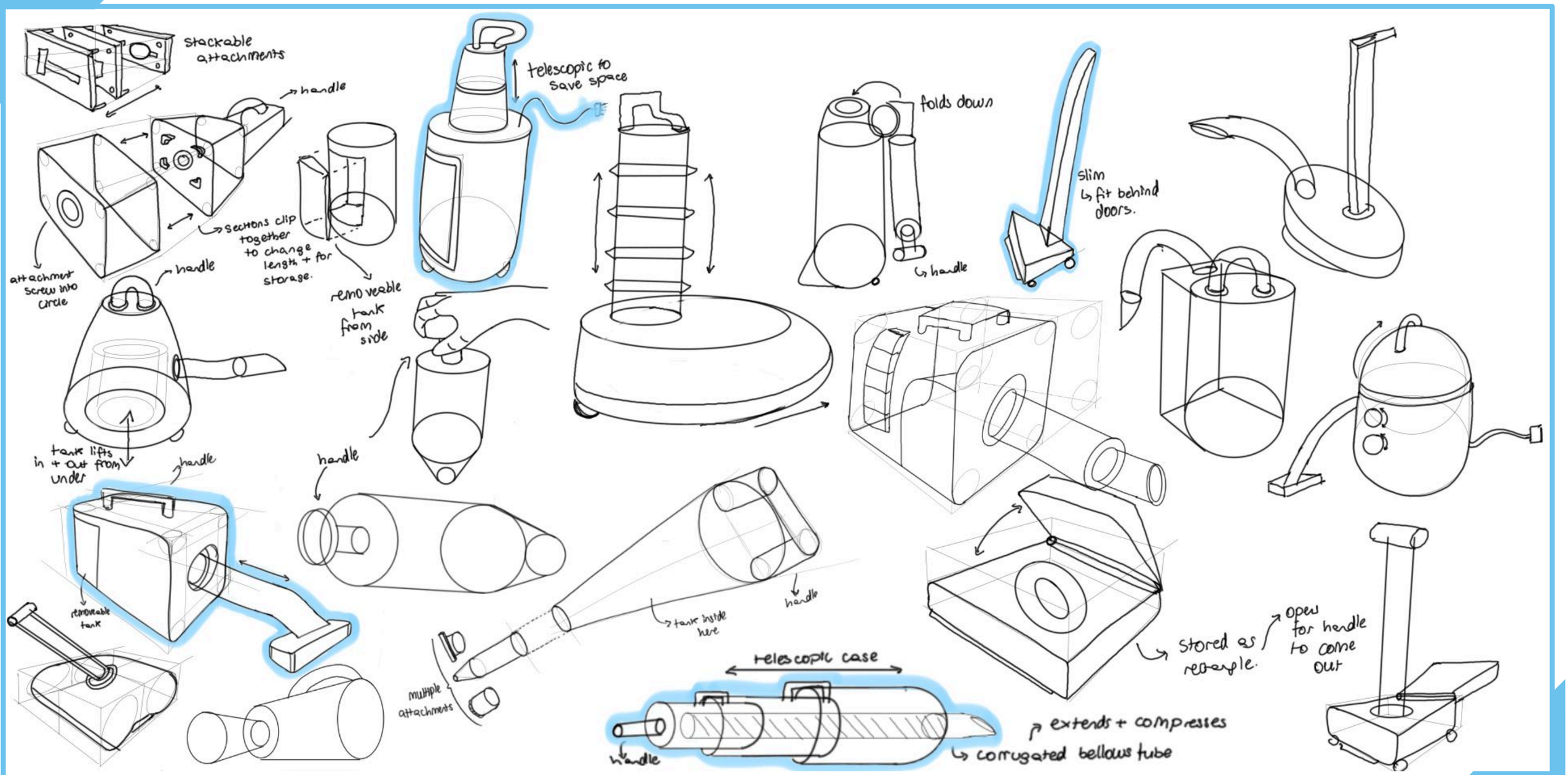
Wet and dry vacuum

Project work: wet and dry vacuum for daycare - ideation



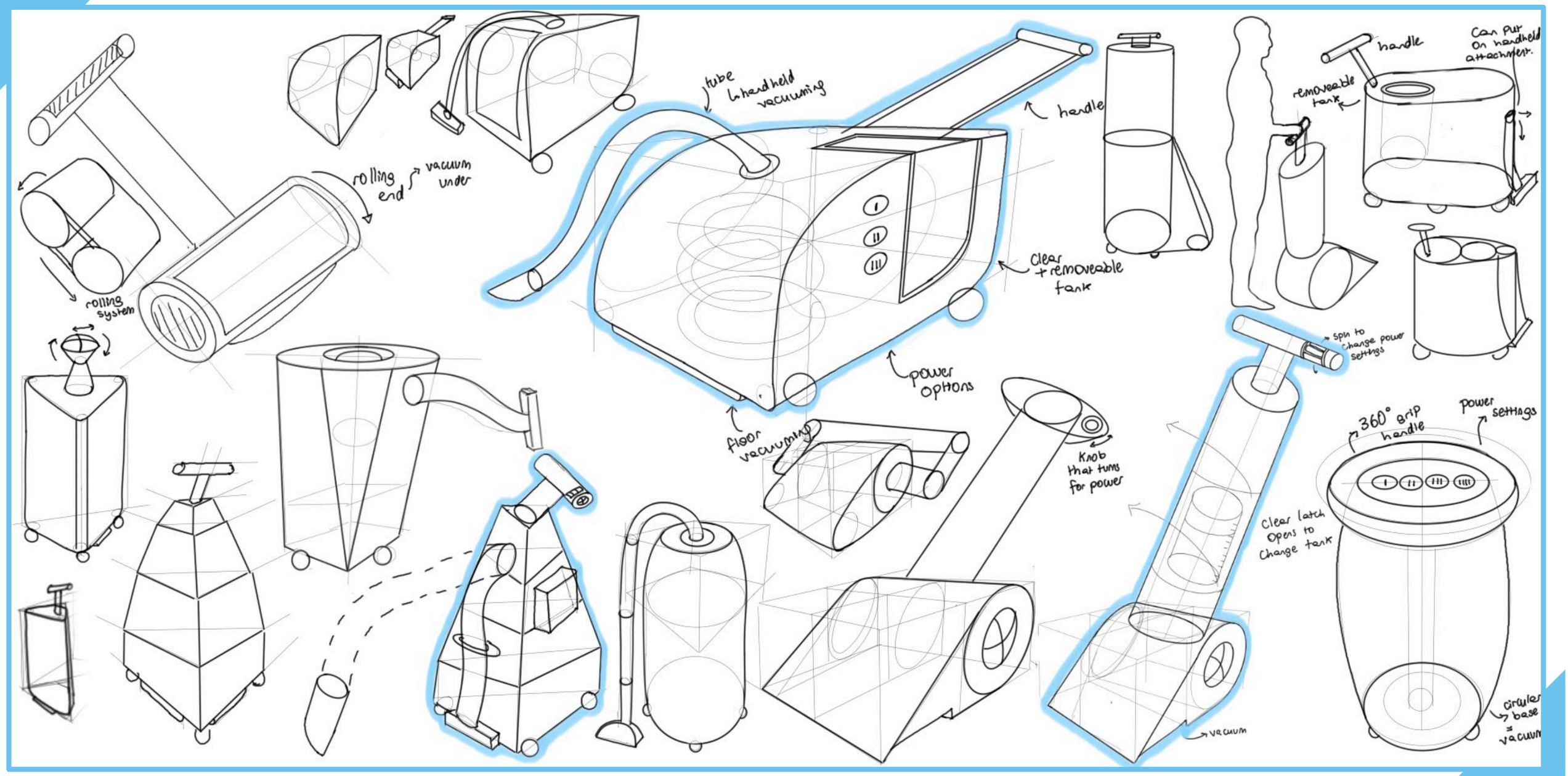
Digital sketching for early stages of ideation for the target market of use in a 'daycare'. The most successful ideas in my opinion are highlighted.

Project work: wet and dry vacuum for students - ideation



Digital sketching for early stages of ideation for the target market of use by 'students'. The most successful ideas in my opinion are highlighted.

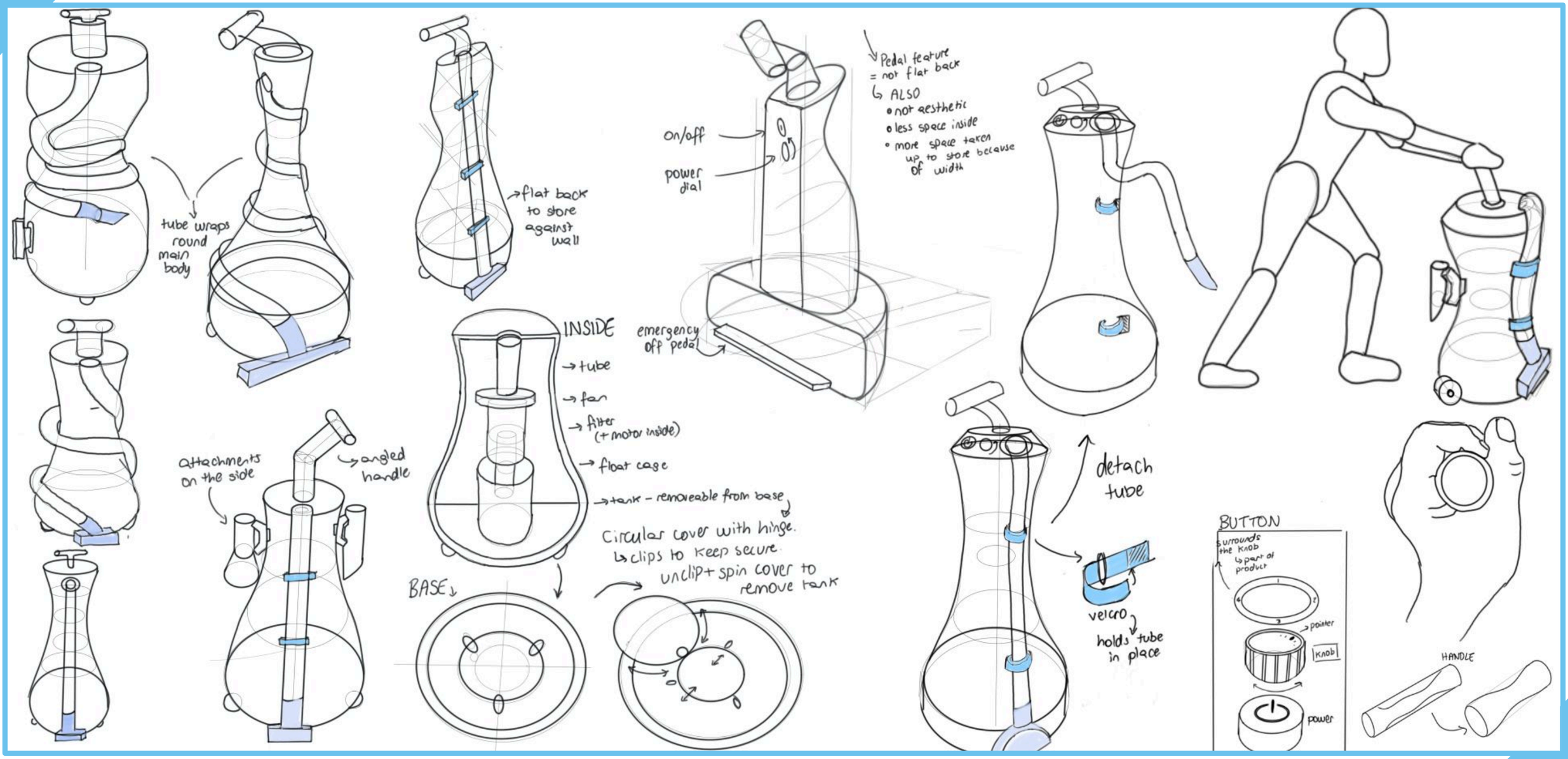
Project work: wet and dry vacuum for care homes - ideation



Digital sketching for early stages of ideation for the target market of use in a 'care home'. The most successful ideas in my opinion are highlighted.

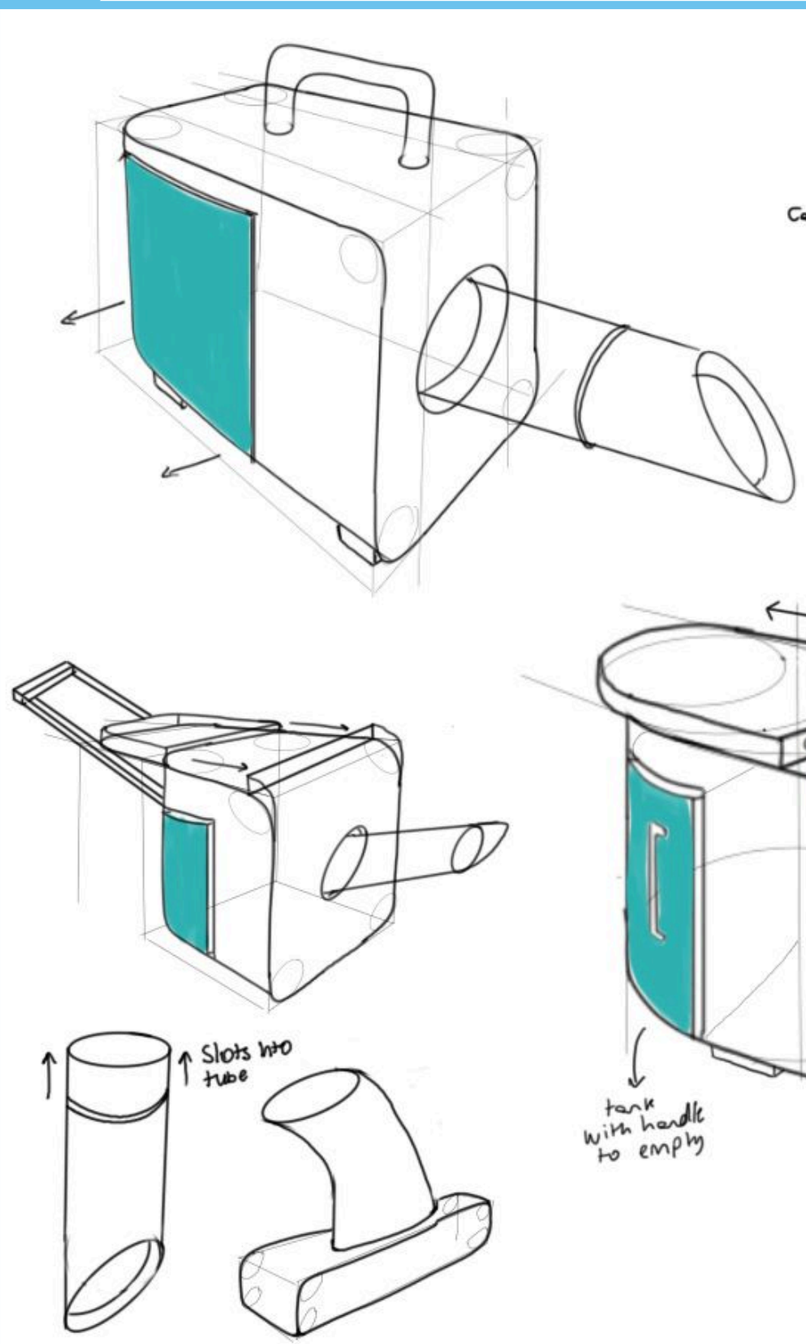
Project work: Wet and dry vacuum - concept development

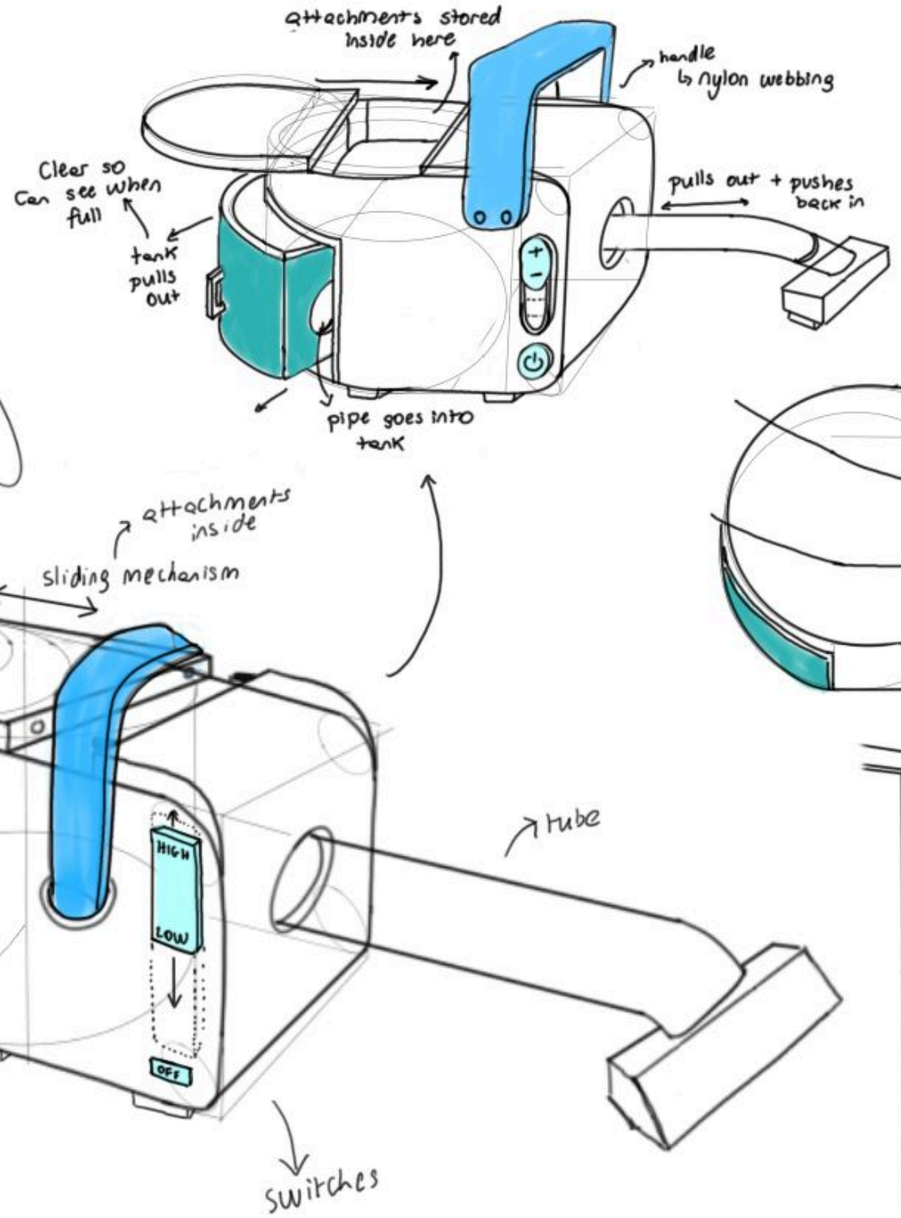
This is my development of a chosen concept for a wet and dry vacuum cleaner in a daycare or childcare environment.

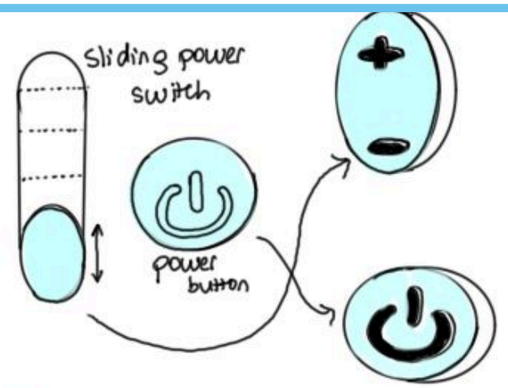


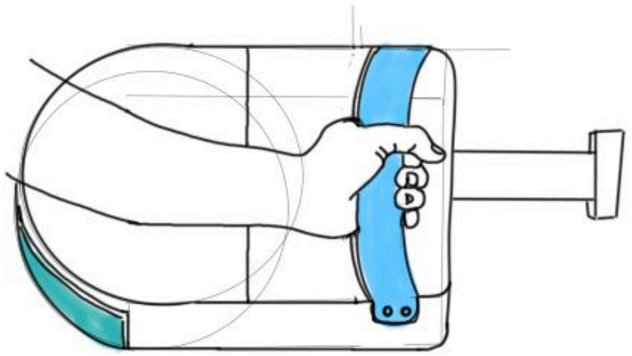
Project work: Wet and dry vacuum - concept development

This is my development of a chosen concept for a wet and dry vacuum cleaner in a student accommodation.





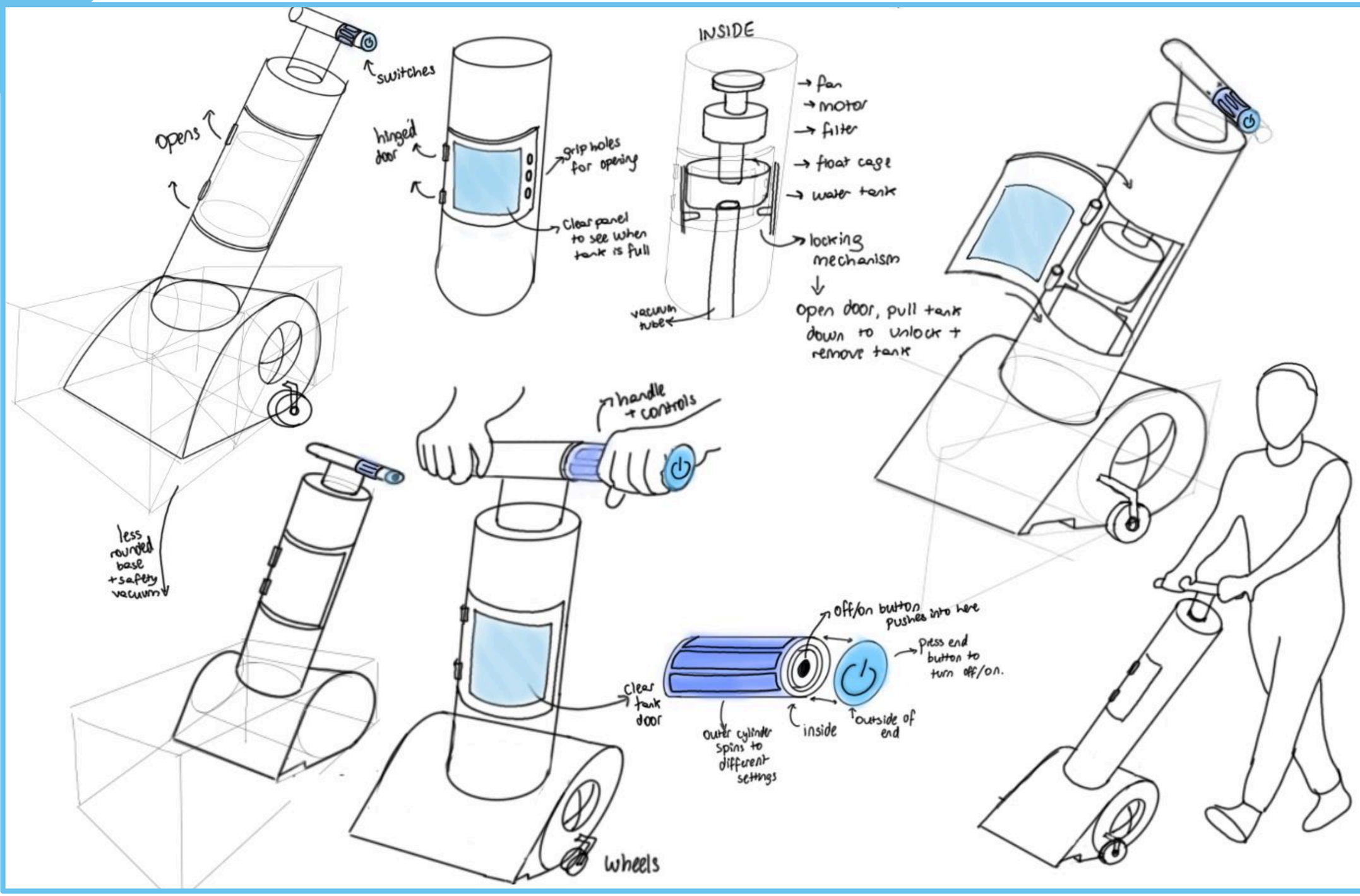


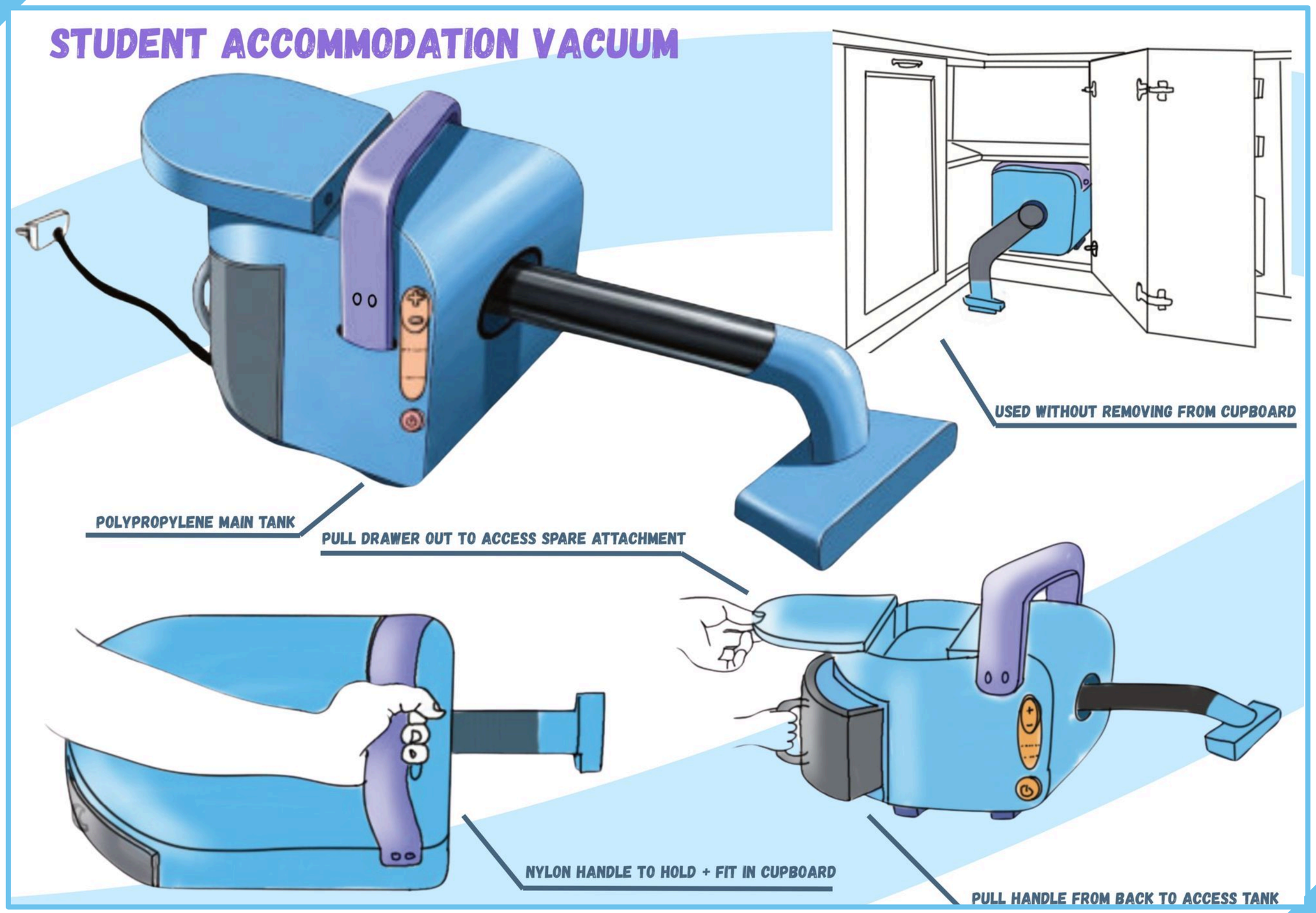


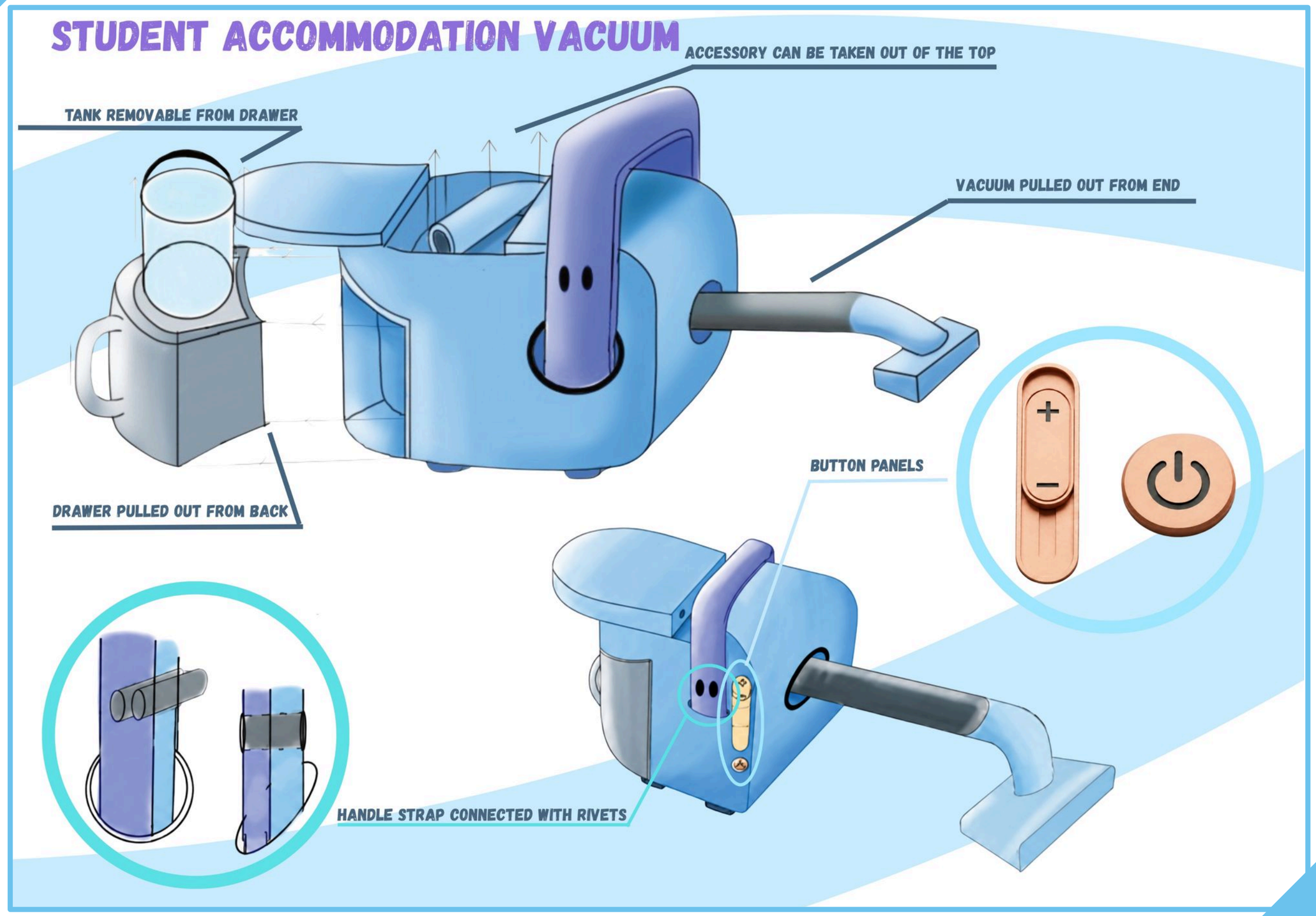


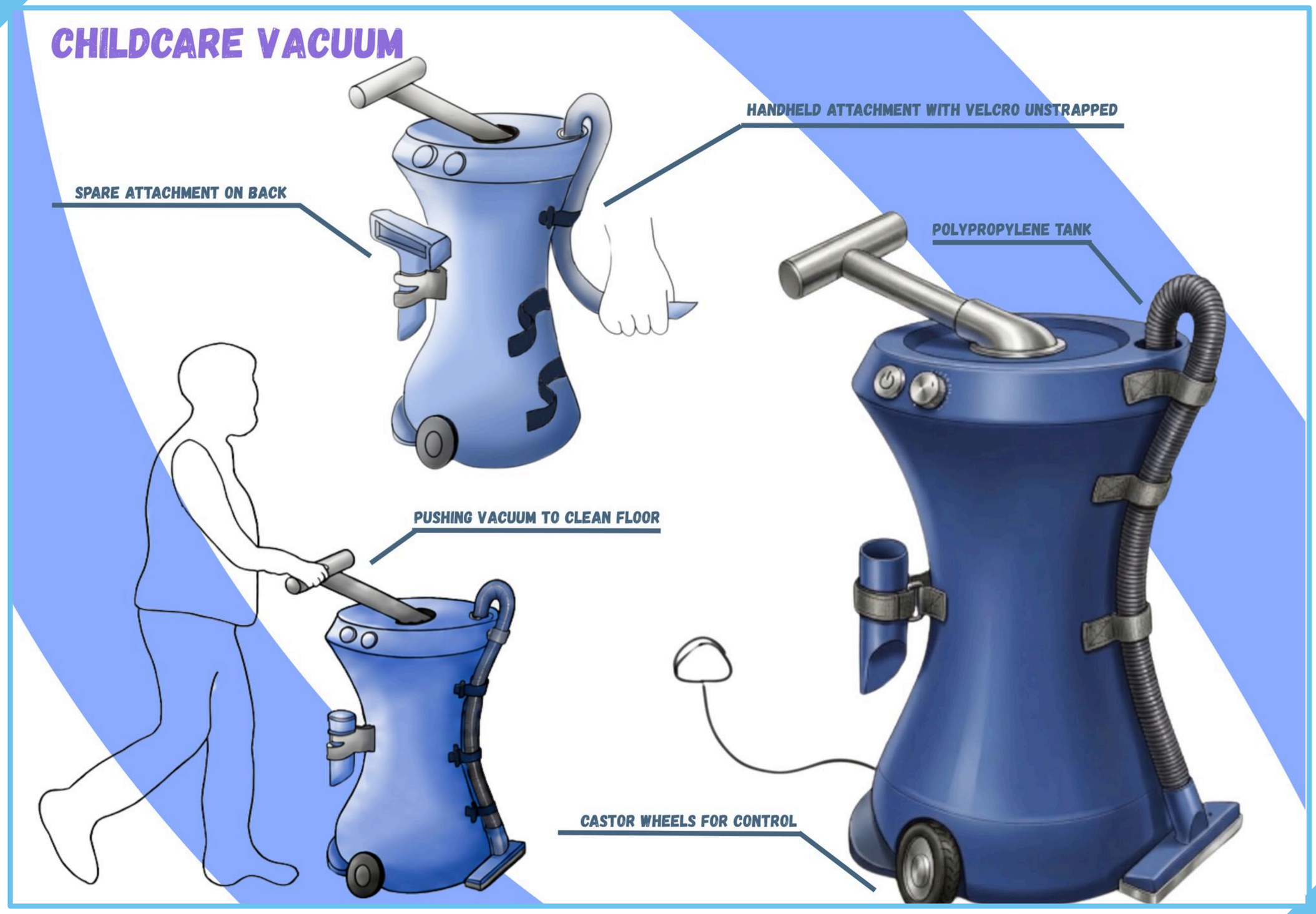
Project work: Wet and dry vacuum - concept development

This is my development of a chosen concept for a wet and dry vacuum cleaner in a care home.

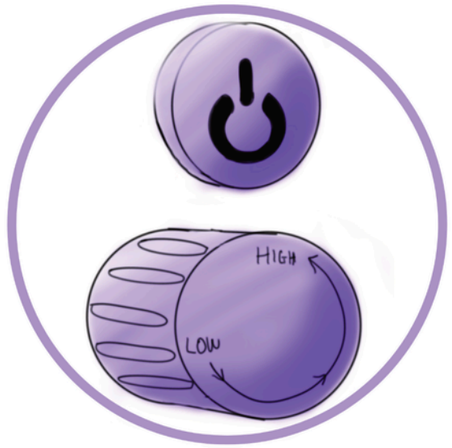



STUDENT ACCOMMODATION VACUUM
USED WITHOUT REMOVING FROM CUPBOARD
POLYPROPYLENE MAIN TANK
PULL DRAWER OUT TO ACCESS SPARE ATTACHMENT
NYLON HANDLE TO HOLD + FIT IN CUPBOARD
PULL HANDLE FROM BACK TO ACCESS TANK

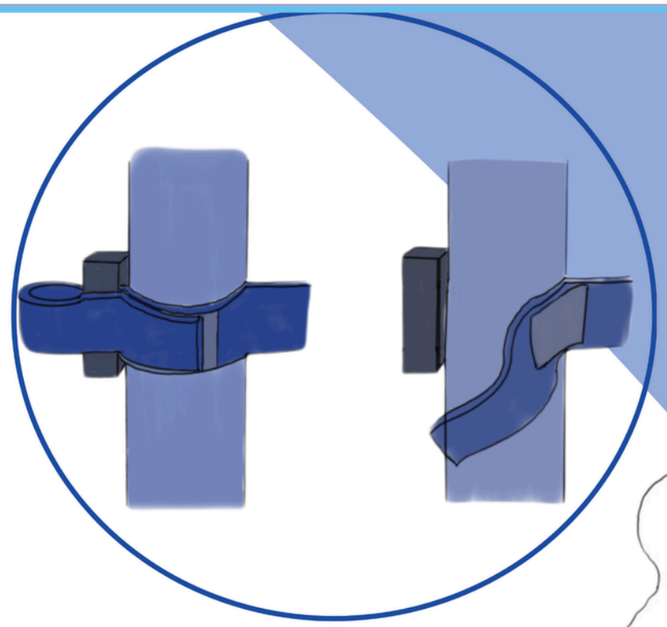

STUDENT ACCOMMODATION VACUUM
ACCESSORY CAN BE TAKEN OUT OF THE TOP
TANK REMOVABLE FROM DRAWER
VACUUM PULLED OUT FROM END
BUTTON PANELS
DRAWER PULLED OUT FROM BACK
HANDLE STRAP CONNECTED WITH RIVETS


CHILDCARE VACUUM
HANDHELD ATTACHMENT WITH VELCRO UNSTRAPPED
SPARE ATTACHMENT ON BACK
POLYPROPYLENE TANK
PUSHING VACUUM TO CLEAN FLOOR
CASTOR WHEELS FOR CONTROL

CHILDCARE VACUUM

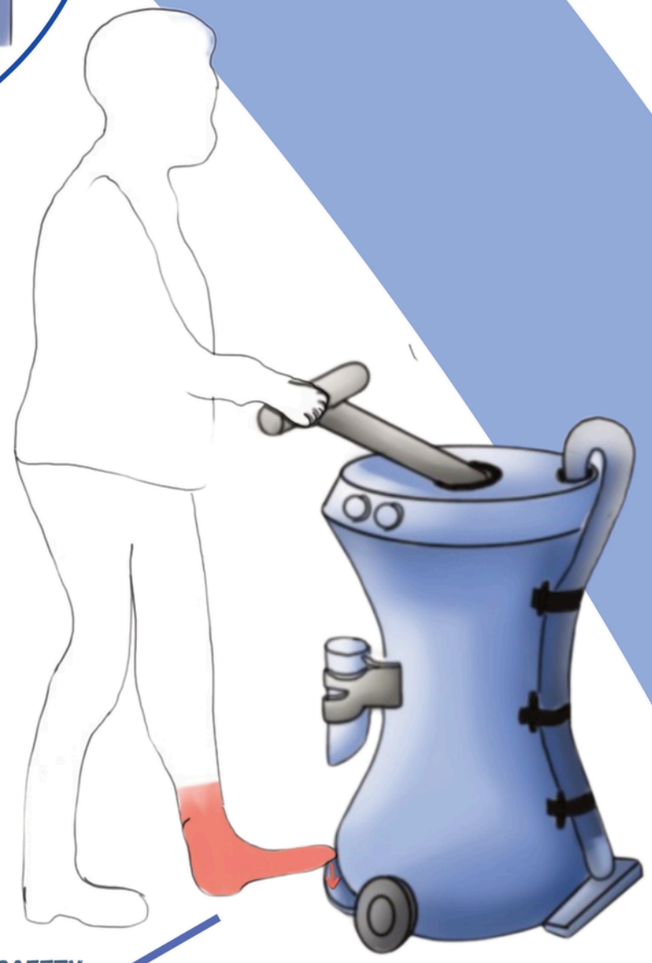


BUTTON AND KNOB TO CHANGE POWER

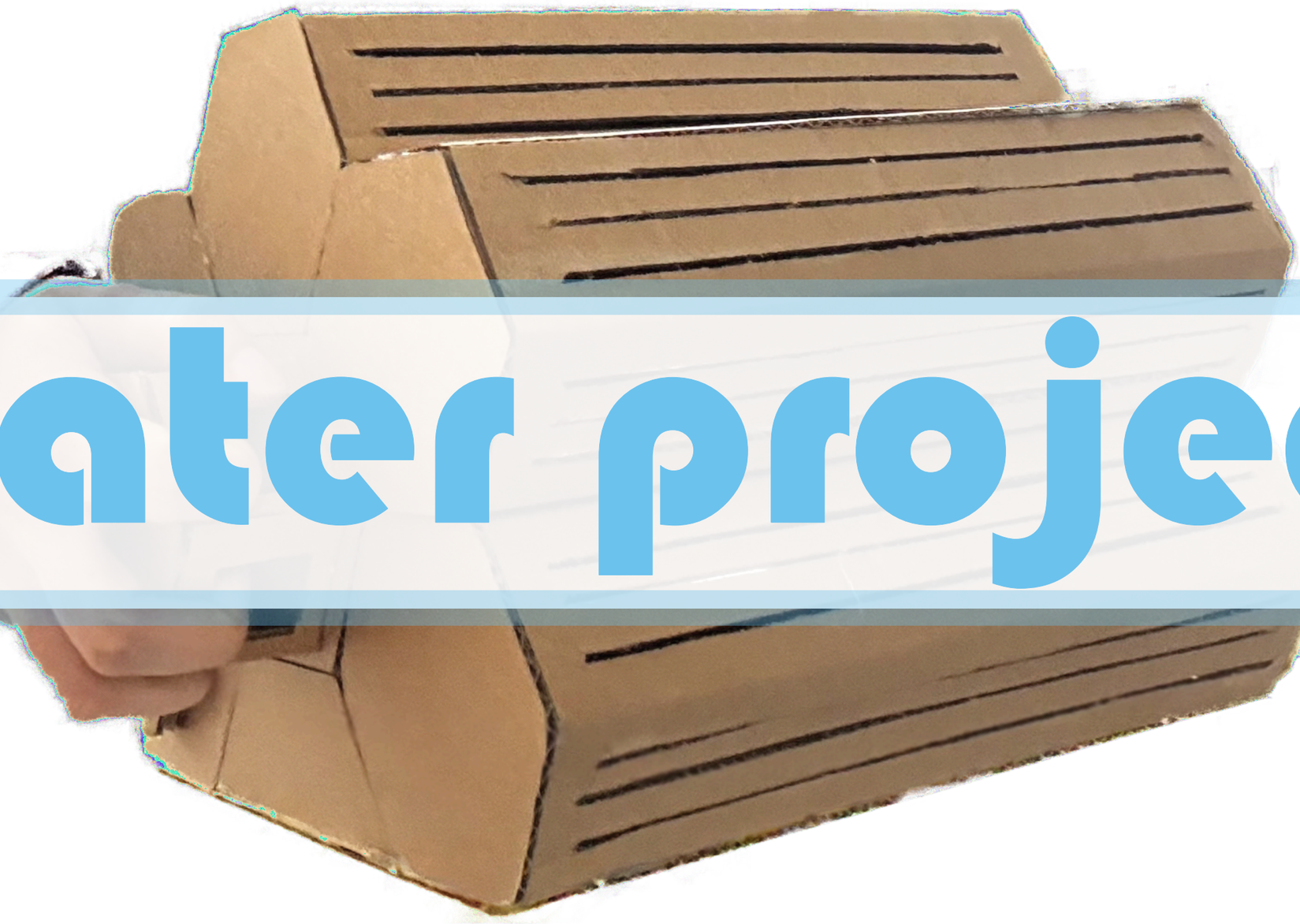


VELCRO STRAPS TO HOLD TUBE IN PLACE



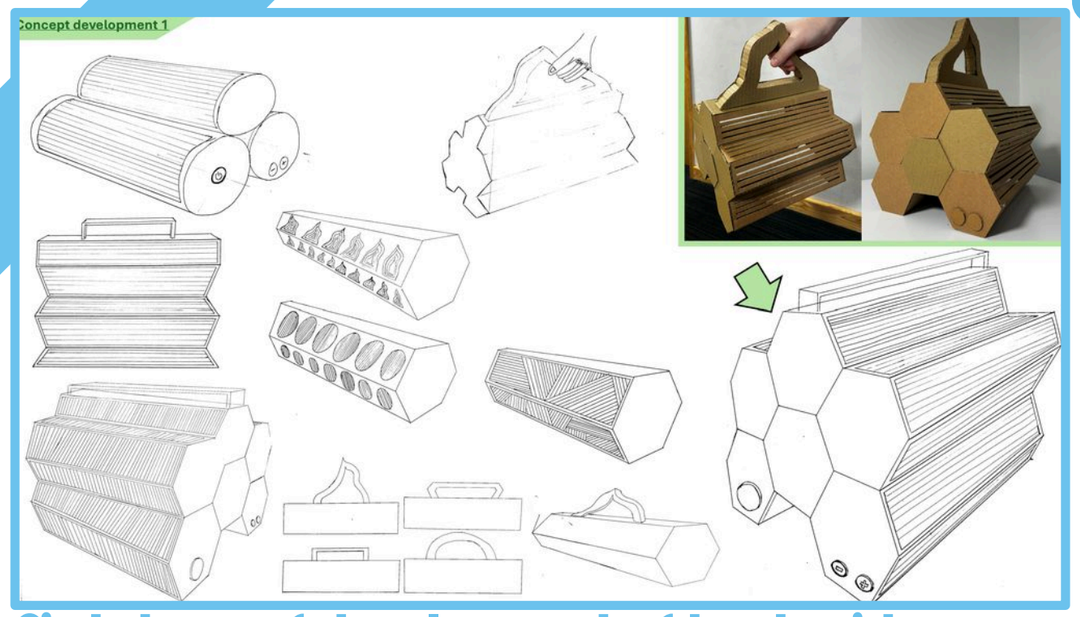


EMERGENCY STOP PEDAL FOR CHILD SAFETY

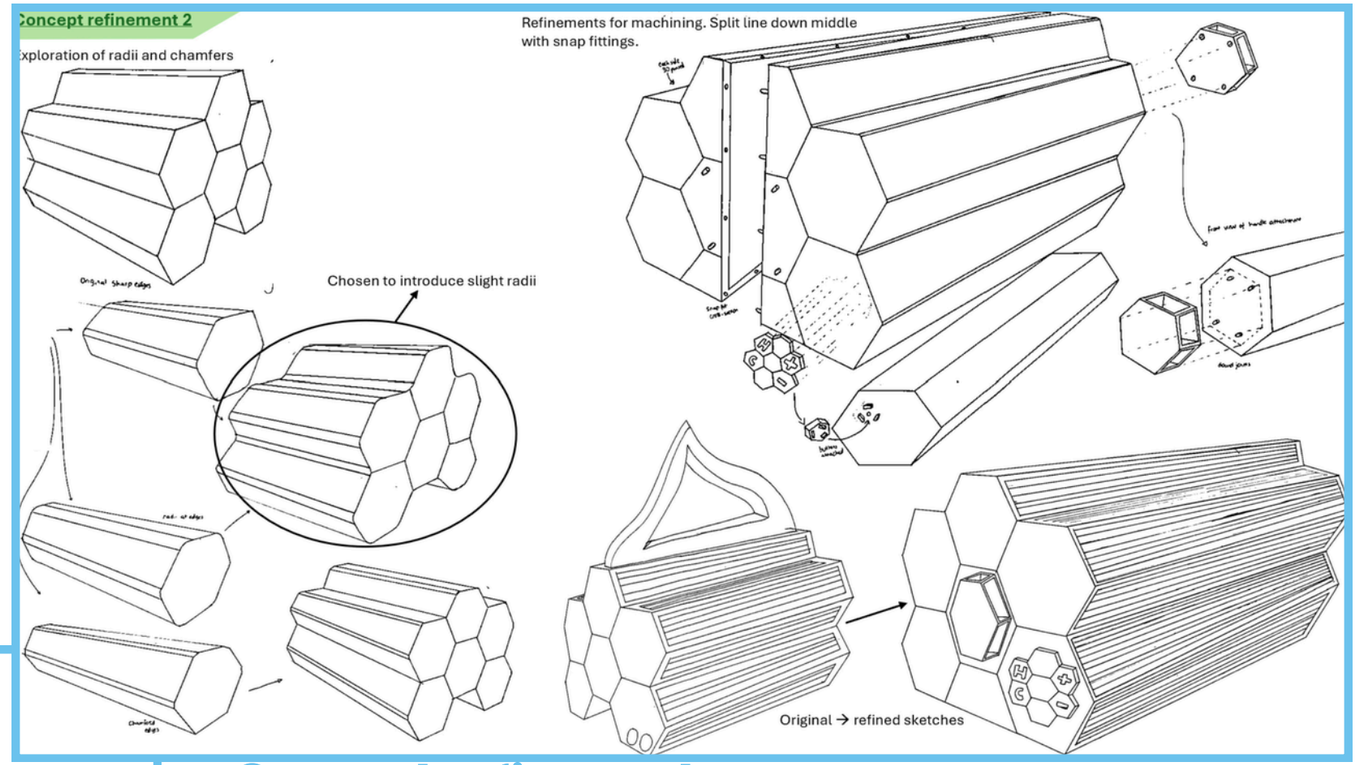


Heater project

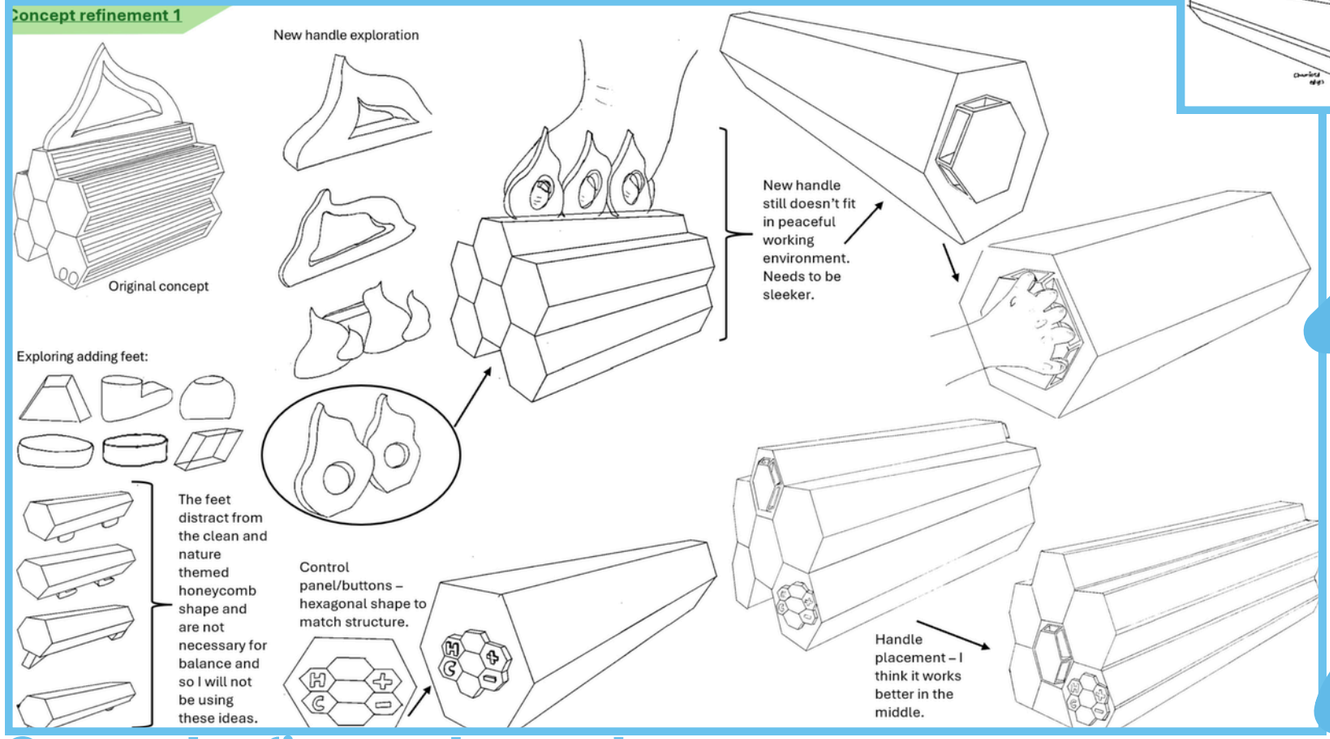
Project work: Heater project - ideation + refinement drawing



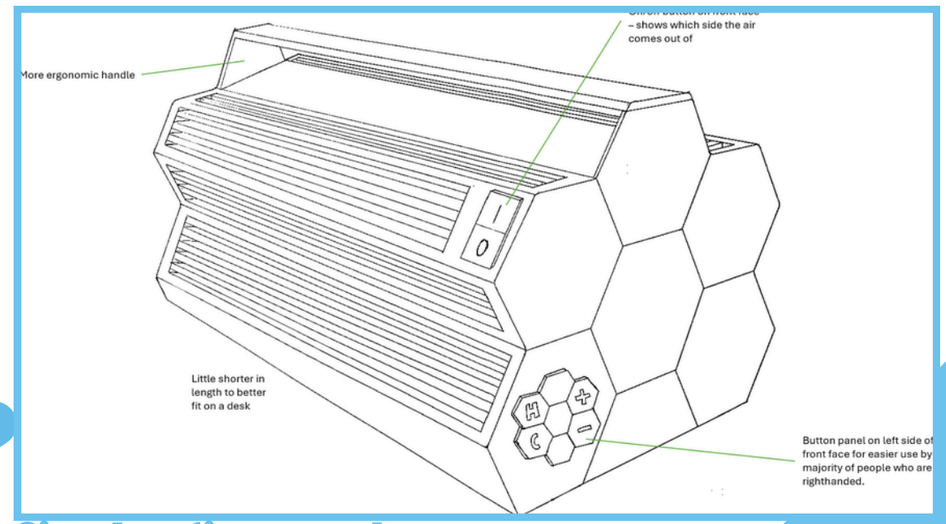
First stage of development of heater idea



Concept refinement page one

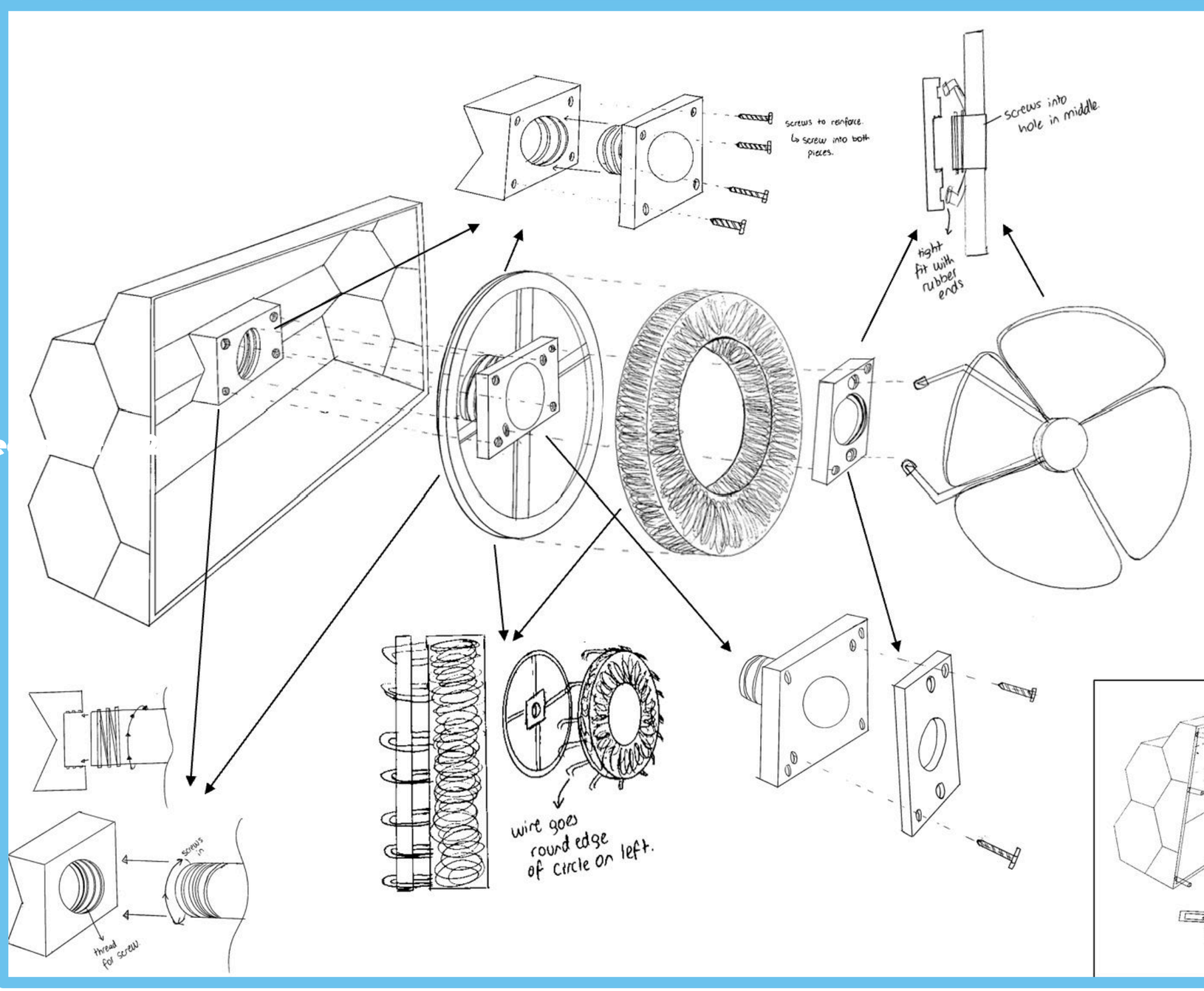


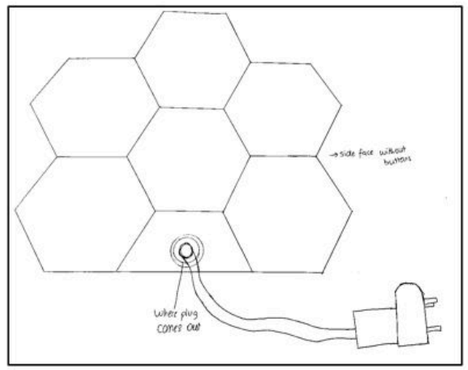
Concept refinement page two

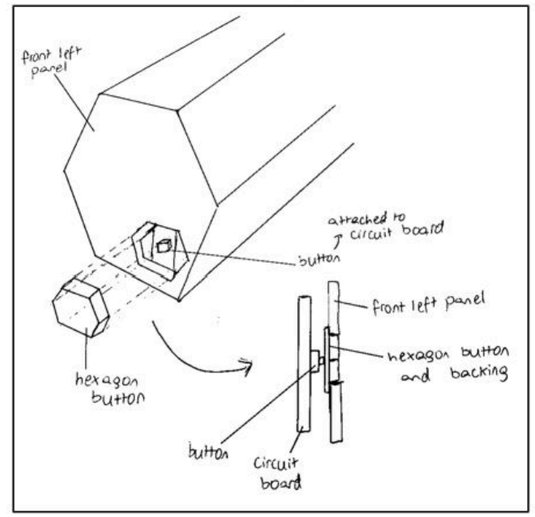


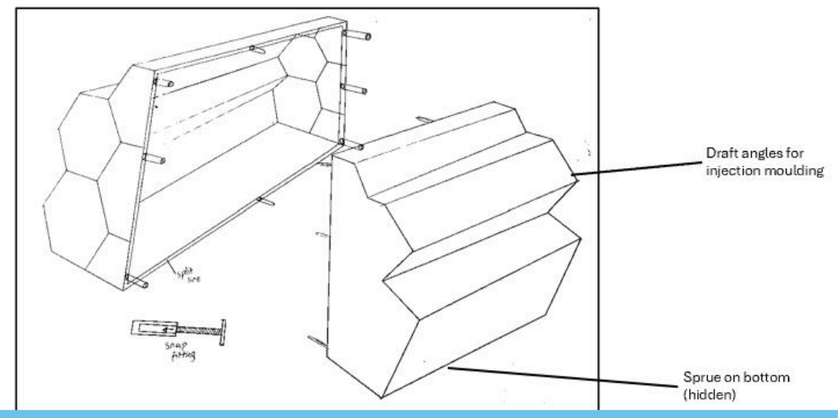
Final refinements

Project work: Heater project - exploded diagram



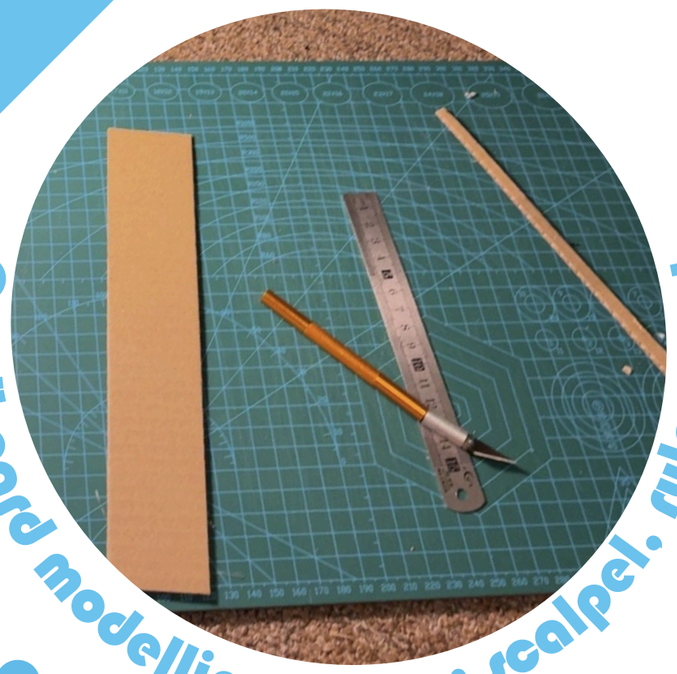




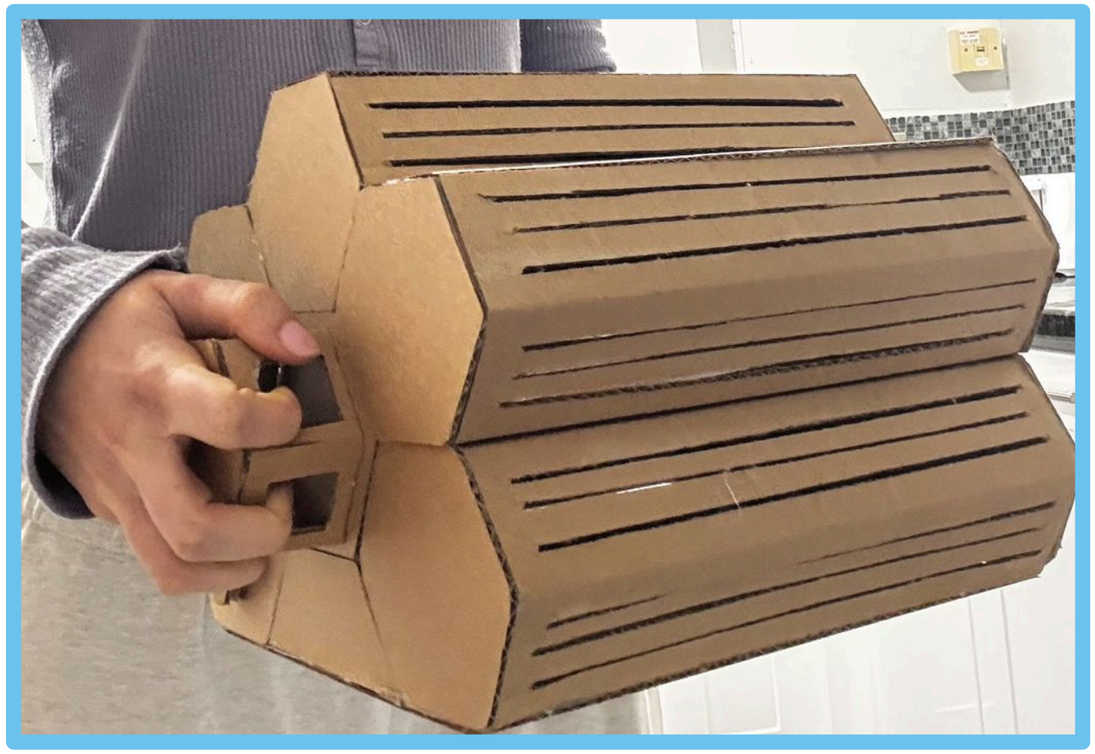


Exploded diagram showing inner workings of the heater and connection method of the parts. Used a heater that I took apart myself to see how the mechanics work.

Project work: Heater project - cardboard modelling



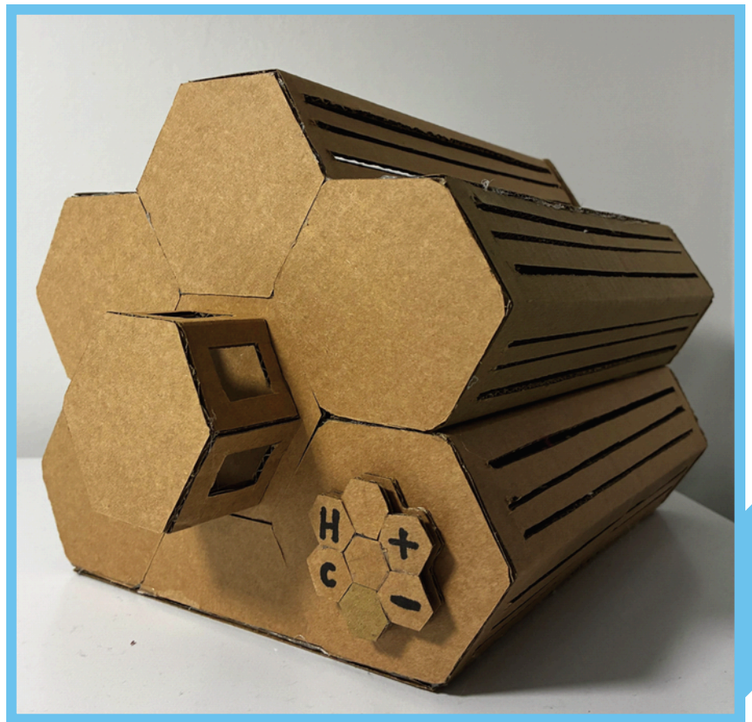
Cardboard modelling set up: scalpel, ruler, mat

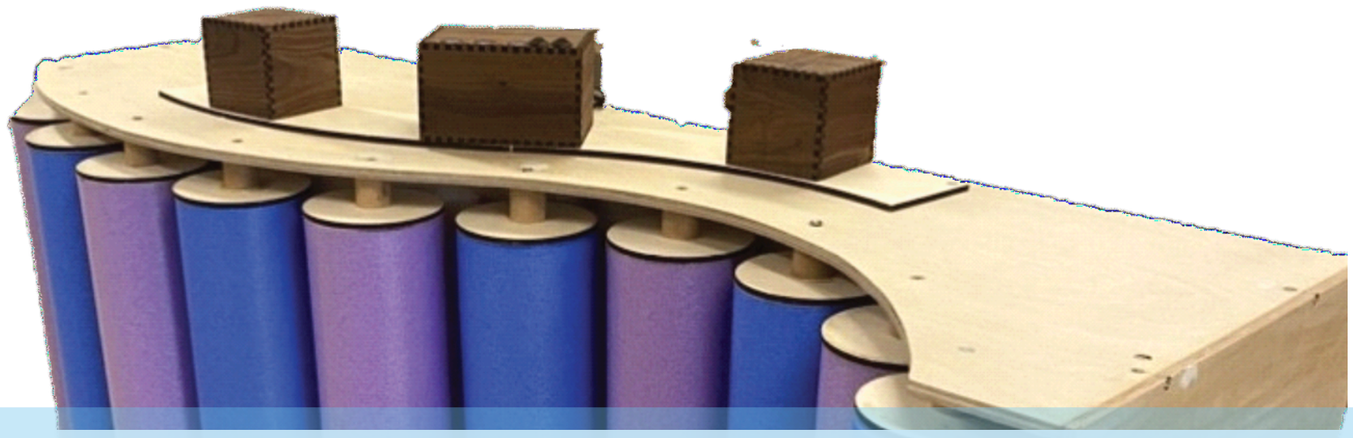


Final cardboard model of heater.
Shows ergonomic usage and detailed buttons.
Demonstrated to me aesthetics in shape and size which is hard to see through drawing.
I liked the soft edges of the product.



All pieces cut using template before gluing



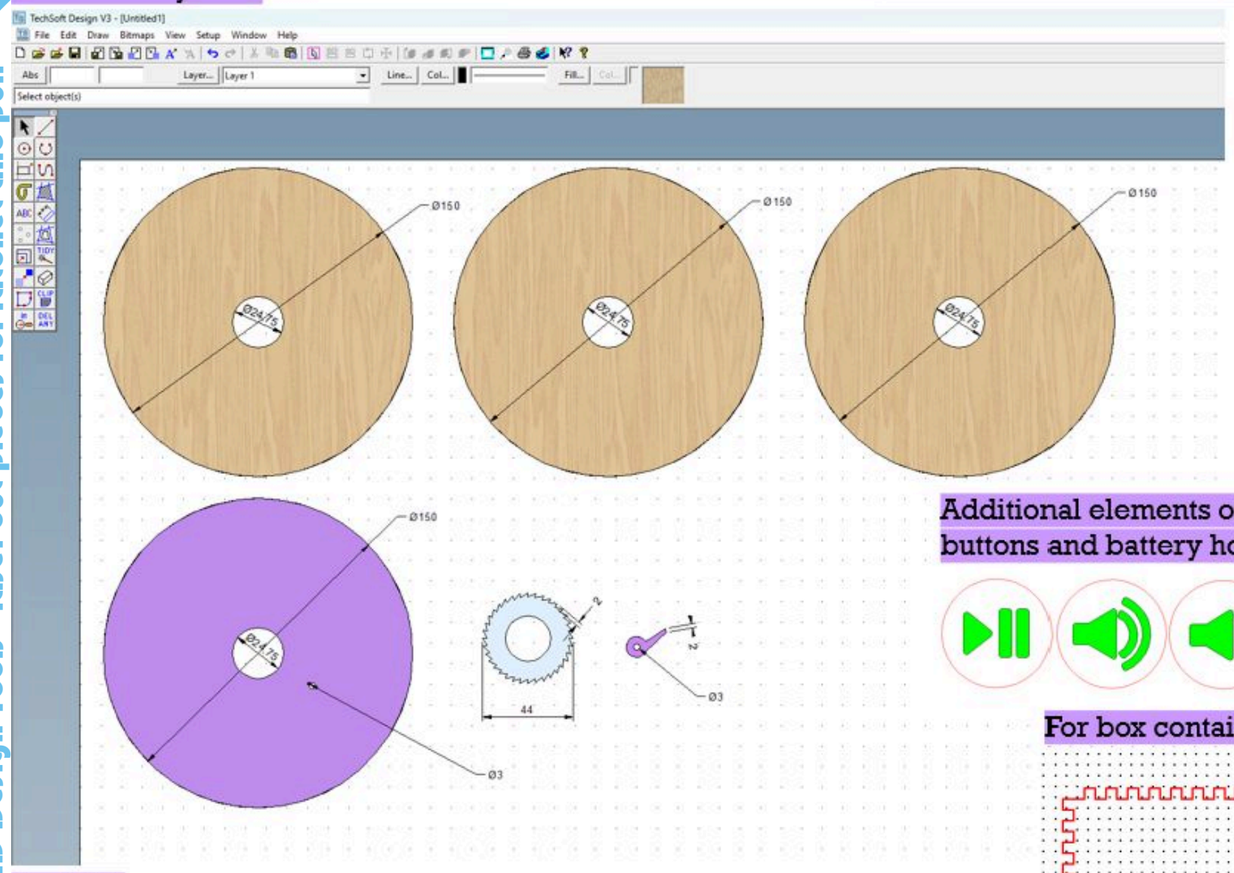


Sensory walkway

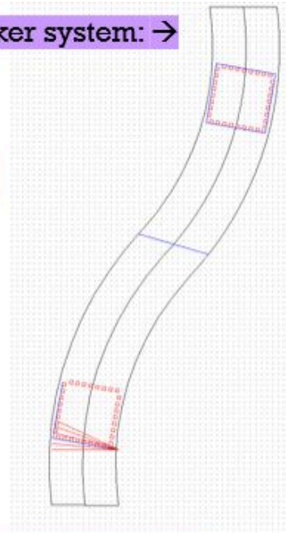
Project work: Sensory walkway - 2D Design Tools + SolidWorks

2D Design Tools - laser cut pieces for ratchet and pull

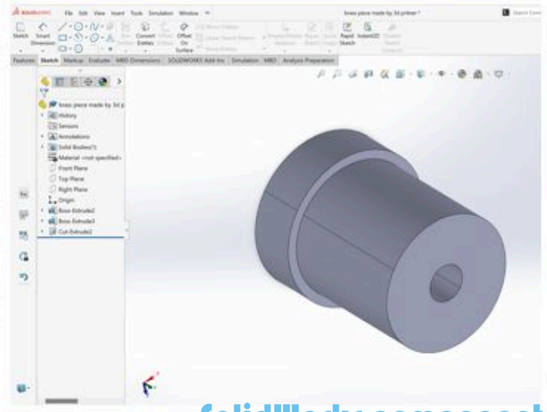
For ratchet system:



For platform holding speaker system: →

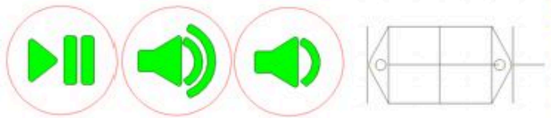


For spinning mechanism – to 3d print:

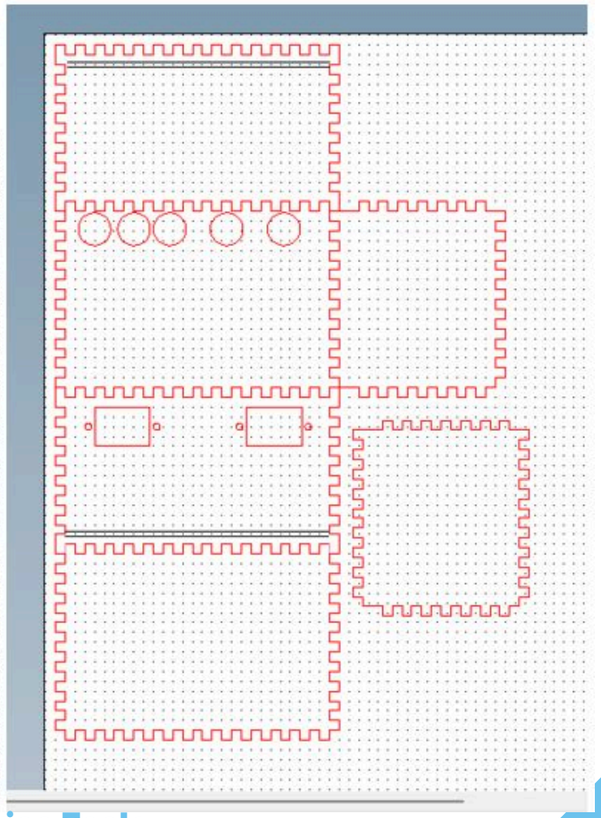


SolidWorks component

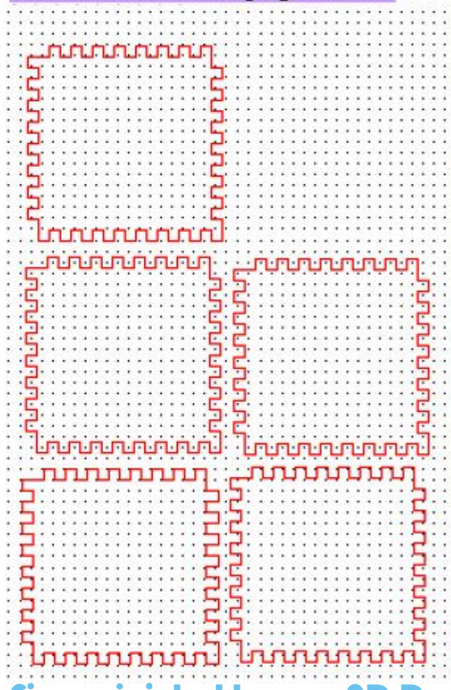
Additional elements of systems box – buttons and battery holder:



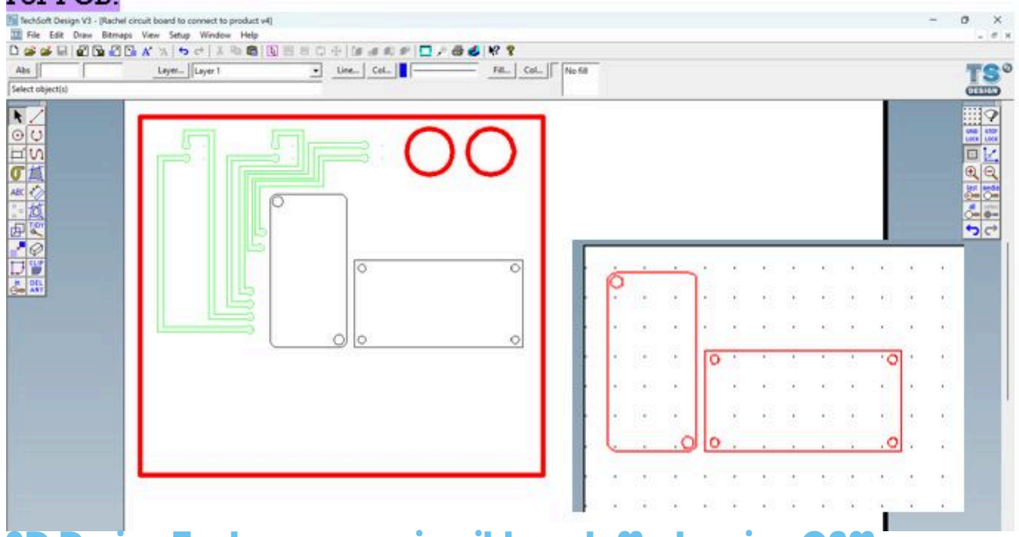
For box containing circuitry: :



For box containing speakers:



For PCB:

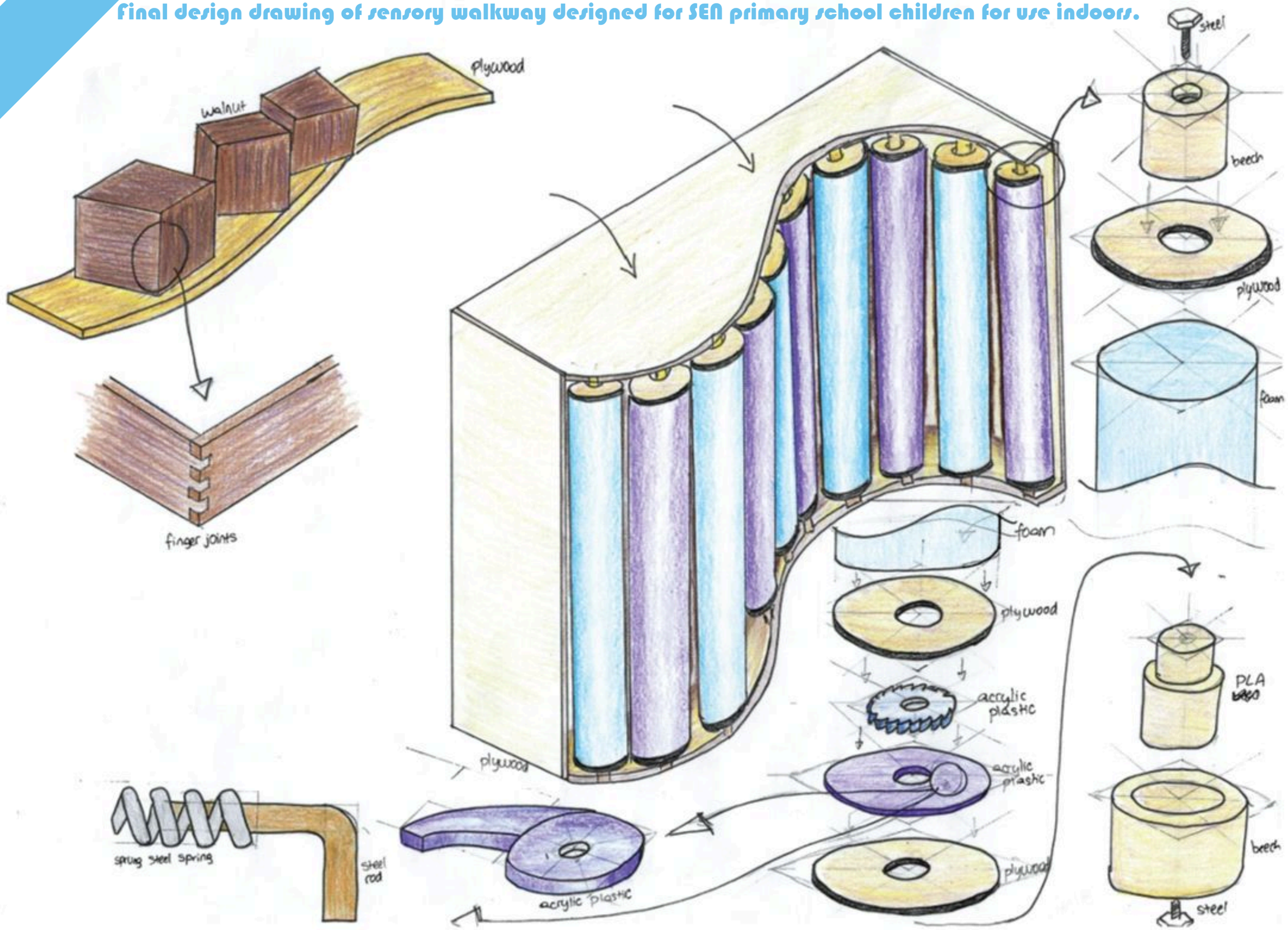


2D Design Tools - copper circuit board. Made using CAM.

Finger-jointed box on 2D Design Tools.

Project work: Sensory walkway - final design drawing

Final design drawing of sensory walkway designed for SEN primary school children for use indoors.



Project work: Sensory walkway - user testing



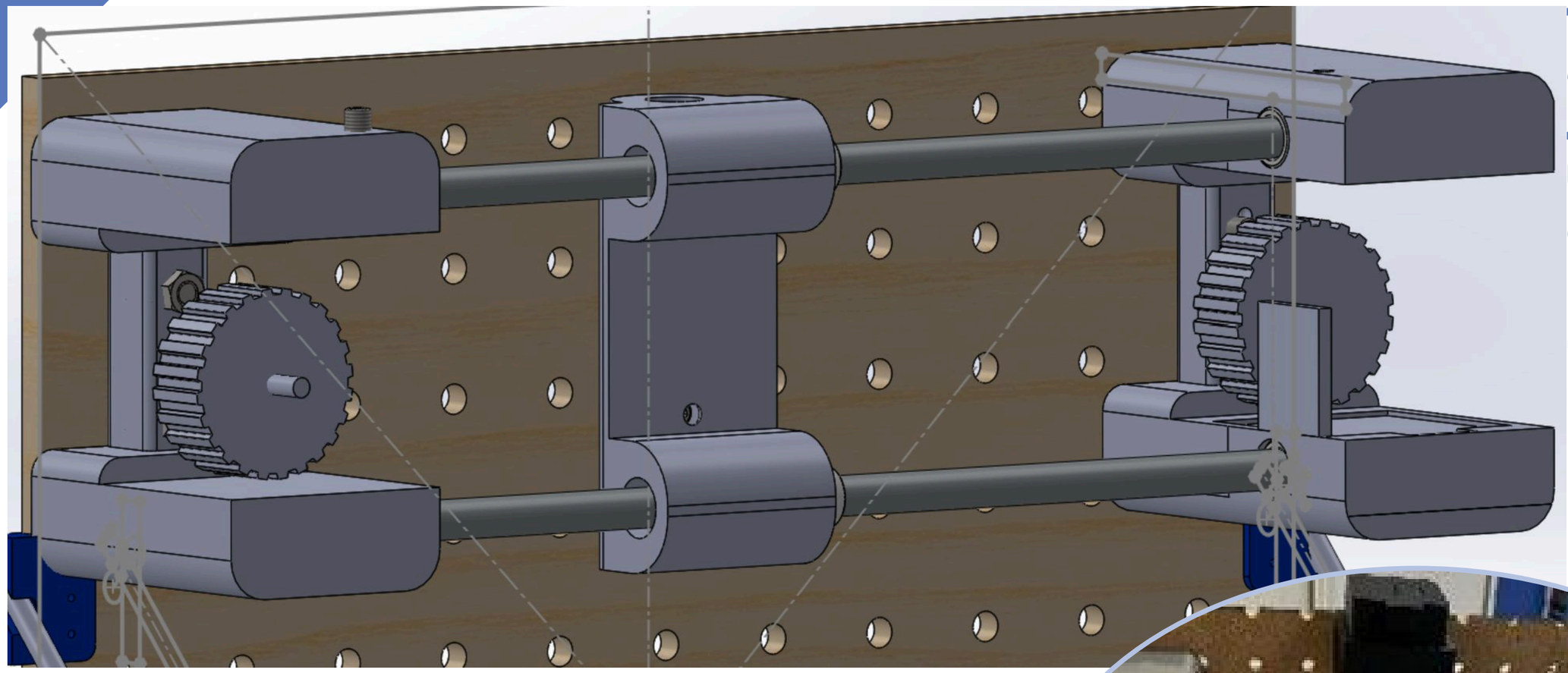




Use of my product by stakeholders. These are children of similar age to those who would use the product. I interviewed them numerous times throughout the design stage, and at the end to evaluate. The product was a success in that they enjoyed interacting with it.

Technical skills

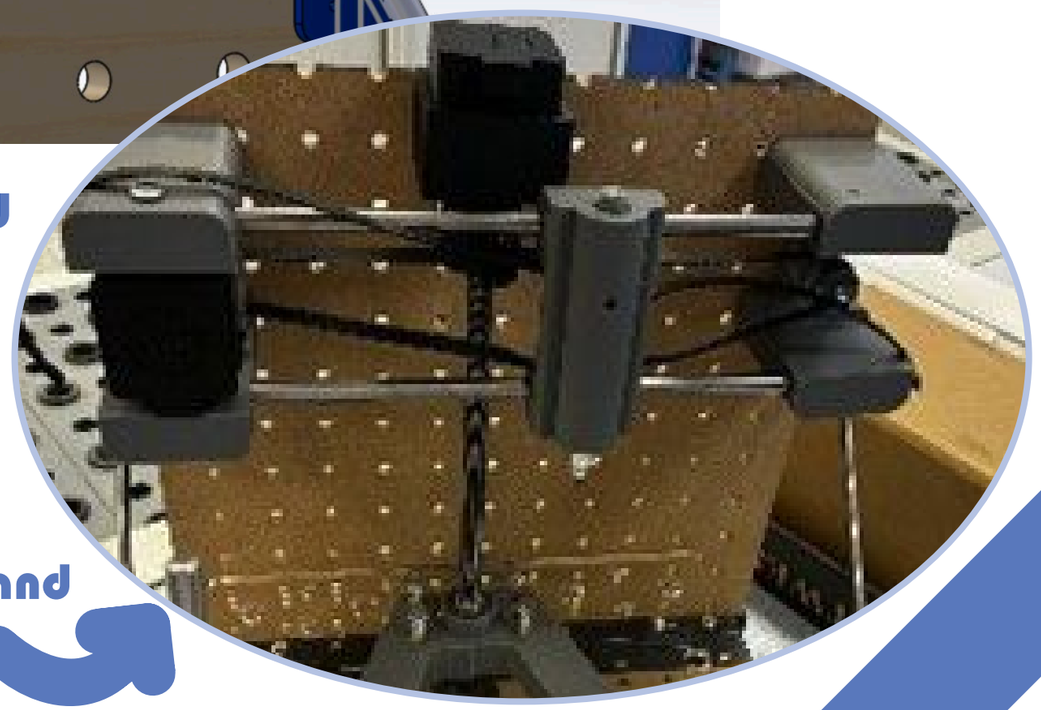
Technical skills: SolidWorks assembly

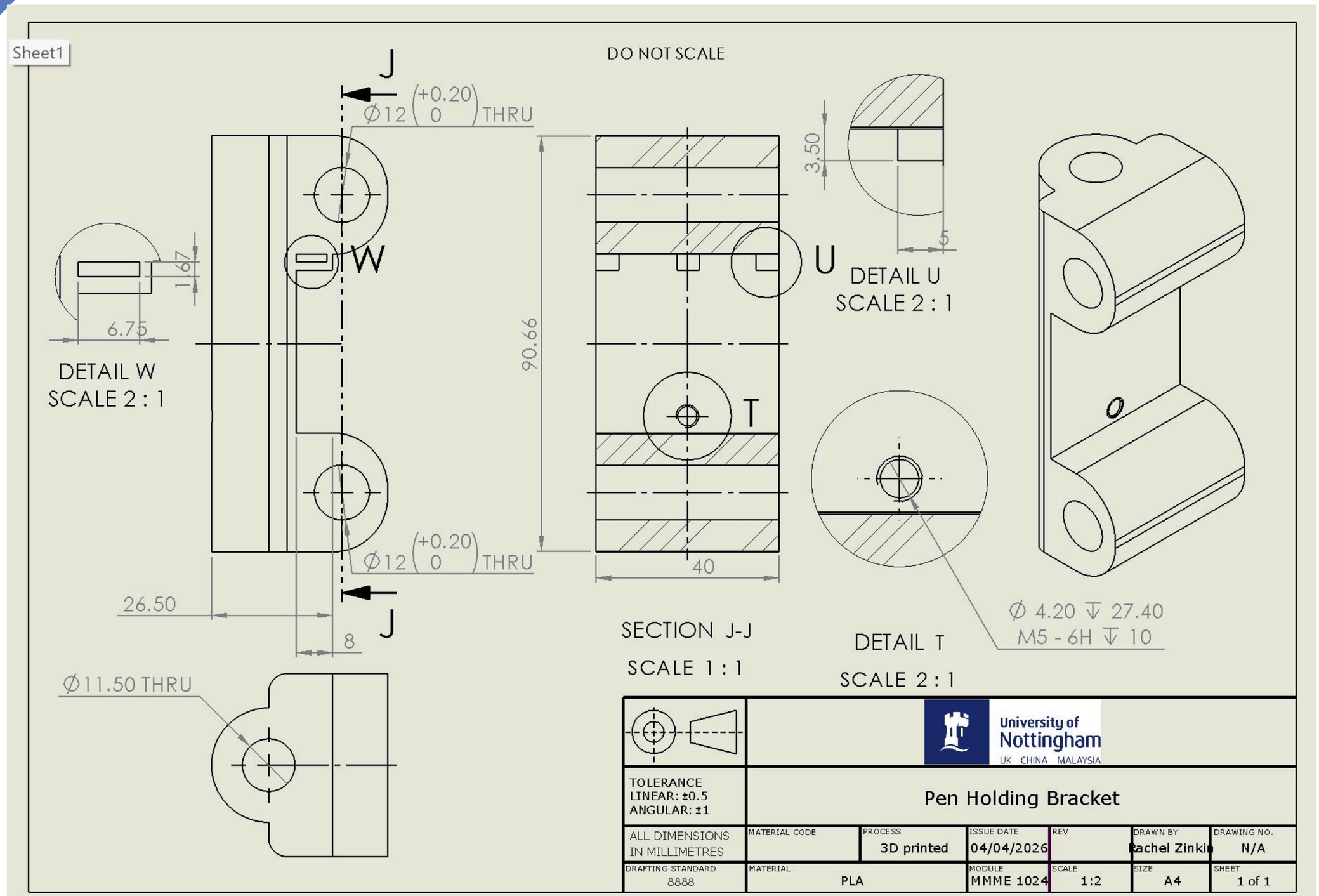


The parts were designed separately and then I used the assembly feature on SolidWorks to put it together.

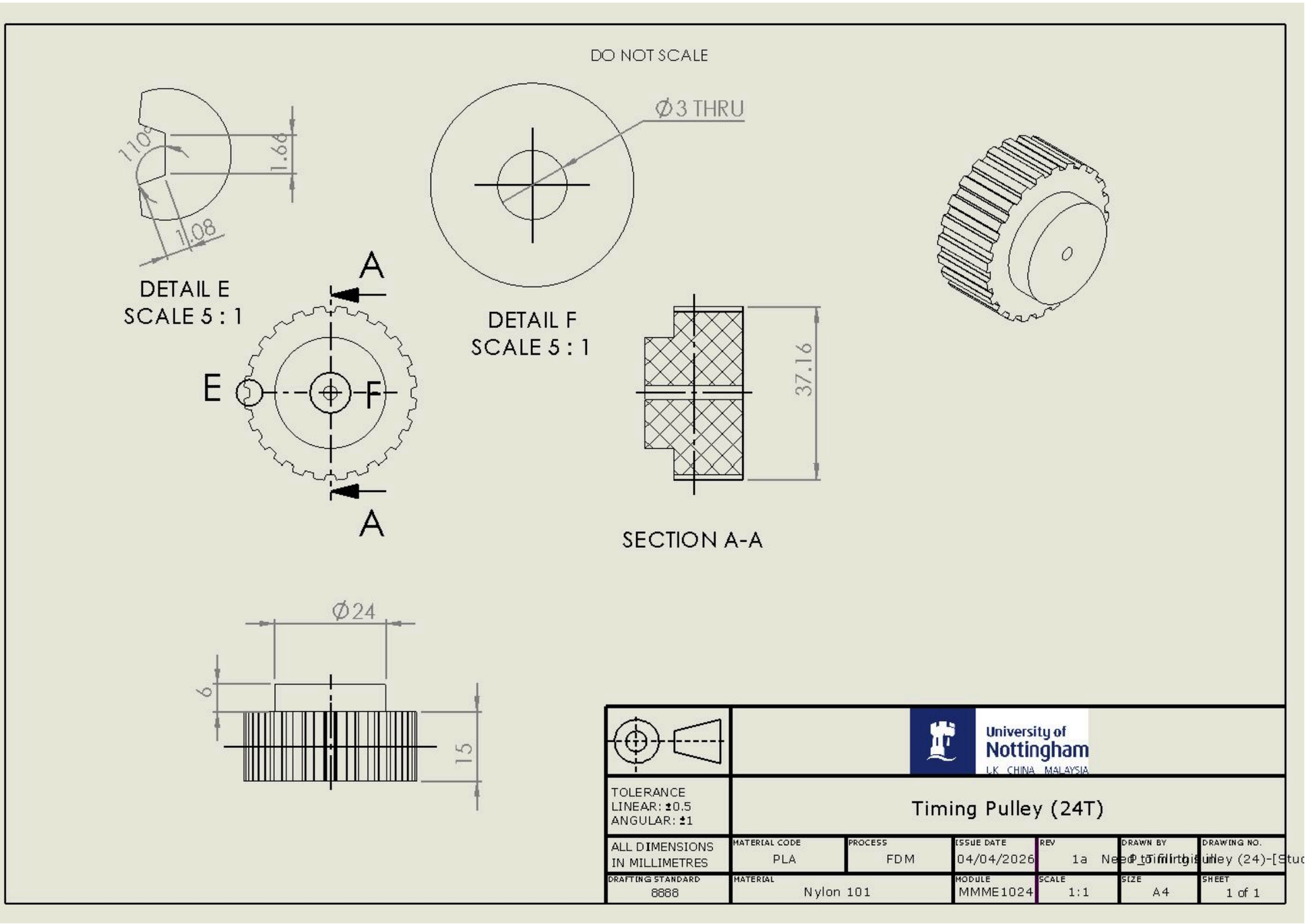
Belt and pulley system to move middle bracket left and right. I designed this using SolidWorks and some standard components provided to me. The three big brackets were all designed by me and my partner. The middle one holds a pen, and as it slides left and right a line is drawn.

The finished system is shown here complete with motors and belt.





Detailed drawing of 3D-printed component of above system.



Detailed drawing of 3D-printed standard component of above system.

Technical skills: Cardboard modelling





Practising cardboard modelling before prototyping my own designs.